

Appendices

This appendix contains additional figures and tables for the thesis. Note that some tables are very small and might be hard to read in the printed version of the thesis. It is recommended to view the tables in the digital version of the thesis, available on GitHub alongside the other resources: <https://github.com/KaiPonel/MRLQuantization>.

Shortening	Task Name
q	Datatype of Embedding (Quantization Technique)
m	Dimensionality of Embedding (Matryoshka size)
l	Inference Layer Number
f32	32-Bit Floating-Point
int	8-Bit Integer
bin	Binary
bits req.	Bits Required to store embedding
CLA-1	Banking77Classification
CLA-2	EmotionClassification
CLA-3	TweetSentimentExtractionClassification
CLA-4	AmazonCounterfactualClassification
CLA-5	MassiveIntentClassification
CLA-6	MassiveScenarioClassification
CLA-7	MTOPDomainClassification
CLA-8	MTOPIntentClassification
CLU-1	ArXivHierarchicalClusteringP2P
CLU-2	ArXivHierarchicalClusteringS2S
CLU-3	BiorxivClusteringP2P.v2
CLU-4	BiorxivClusteringS2S.v2
CLU-5	MedrxivClusteringP2P.v2
CLU-6	MedrxivClusteringS2S.v2
CLU-7	RedditClustering.v2
CLU-8	StackExchangeClustering.v2
CLU-9	StackExchangeClusteringP2P.v2
CLU-10	TwentyNewsgroupsClustering.v2
STS-1	BIOSSES
STS-2	SICK-R
STS-3	STS12
STS-4	STS13
STS-5	STS14
STS-6	STS15
STS-7	STS16
STS-8	STSBenchmark
STS-9	STS17
STS-10	STS22
PCL-1	SprintDuplicateQuestions
PCL-2	TwitterSemEval2015
PCL-3	TwitterURLCorpus
RET-1	ArguAna
RET-2	CQADupstackWebmastersRetrieval
RET-3	NFCorpus
RER-1	AskUbuntuDupQuestions
RER-2	MindSmallReranking
RER-3	StackOverflowDupQuestions
SUM-1	SummEval

Table 2: Shortening and Task Name Mapping

A. Model Selection

Table 3: Top-100 models on the MTEB leaderboard as of 03.08.2024 at 12:00pm. Model size is measured in million parameters. Dimensions refers to the default embedding size of the model.

Rank	Model	Model Size	Memory Usage	Dimensions	Max Tokens	Avg. Score
1	bge-en-icl	7111	26.49	4096	32768	71.67
2	stella_en_1.5B_v5	1543	5.75	8192	131072	71.19
3	SFR-Embedding-2_R	7111	26.49	4096	32768	70.31
4	gte-Qwen2-7B-instruct-Q4_K_M-GGUF					70.24
5	gte-Qwen2-7B-instruct	7613	28.36	3584	131072	70.24
6	stella_en_400M_v5	435	1.62	8192	8192	70.11
7	bge-multilingual-gemma2	9242	34.43	3584	8192	69.88
8	NV-Embed-v1	7851	29.25	4096	32768	69.32
9	voyage-large-2-instruct			1024	16000	68.23
10	Linq-Embed-Mistral	7111	26.49	4096	32768	68.17
11	SFR-Embedding-Mistral	7111	26.49	4096	32768	67.56
12	gte-Qwen1.5-7B-instruct	7099	26.45	4096	32768	67.34
13	gte-Qwen2-1.5B-instruct-Q4_0-GGUF					67.16
14	gte-Qwen2-1.5B-instruct	1776	6.62	1536	131072	67.16
15	voyage-lite-02-instruct	1220	4.54	1024	4000	67.13
16	GritLM-7B	7242	26.98	4096	32768	66.76
17	e5-mistral-7b-instruct	7111	26.49	4096	32768	66.63
18	google-gecko.text-embedding-preview-0409	1200	4.47	768	2048	66.31
19	TDTE					65.96
20	GritLM-8x7B	46703	173.98	4096	32768	65.66
21	gte-large-en-v1.5	434	1.62	1024	8192	65.39
22	LLM2Vec-Meta-Llama-3-supervised	7505	27.96	4096	8192	65.01
23	LLM2Vec-Mistral-supervised	7111	26.49	4096	32768	64.8
24	echo-mistral-7b-instruct-lasttoken	7111	26.49	4096	32768	64.68

Continued on next page

Rank	Model	Model Size	Memory Usage	Dimensions	Max Tokens	Avg. Score
25	mxbai-embed-large-v1	335	1.25	1024	512	64.68
26	UAE-Large-V1	335	1.25	1024	512	64.64
27	text-embedding-3-large			3072	8191	64.59
28	voyage-lite-01-instruct			1024	4000	64.49
29	Cohere-embed-english-v3.0			1024	512	64.47
30	multilingual-e5-large-instruct	560	2.09	1024	514	64.41
31	google-gecko-256.text-embedding-preview-0409	1200	4.47	256	2048	64.37
32	GIST-large-Embedding-v0	335	1.25	1024	512	64.34
33	bge-large-en-v1.5	335	1.25	1024	512	64.23
34	blade-embed	335	1.25	1024	512	64.21
35	MUG-B-1.6	335	1.25	1024	512	64.2
36	LLM2Vec-Llama-2-supervised	6607	24.61	4096	4096	64.14
37	gte-base-en-v1.5	137	0.51	768	8192	64.11
38	Cohere-embed-multilingual-v3.0			1024	512	64.01
39	GIST-Embedding-v0	109	0.41	768	512	63.71
40	bge-base-en-v1.5	438	1.63	768	512	63.56
41	privacy_embedding_rag_10k_base_final	109	0.41	768	512	63.55
42	privacy_embedding_rag_10k_base_12_final	109	0.41	768	512	63.55
43	privacy_embedding_rag_10k_base_15_final	109	0.41	768	512	63.55
44	privacy_embedding_rag_10k_base_checkpoint_2	109	0.41	768	512	63.55
45	ember-v1	335	1.25	1024	512	63.54
46	sf_model_e5	335	1.25	1024	512	63.34
47	mxbai-embed-2d-large-v1	335	1.25	1024	512	63.25
48	gte-large	335	1.25	1024	512	63.13
49	NoInstruct-small-Embedding-v0	33	0.12	384	512	63.12
50	GIST-small-Embedding-v0	33	0.12	384	512	62.72
51	stella-base-en-v2	55	0.2	768	512	62.61
52	gte-base	109	0.41	768	512	62.39
53	UniVaR-lambda-80	137	0.51	768	8192	62.39

Continued on next page

Rank	Model	Model Size	Memory Usage	Dimensions	Max Tokens	Avg. Score
54	UniVaR-lambda-20	137	0.51	768	8192	62.39
55	UniVaR-lambda-1	137	0.51	768	8192	62.39
56	nomi-embed-text-v1	137	0.51	768	8192	62.39
57	UniVaR-lambda-5	137	0.51	768	8192	62.39
58	nomi-embed-text-v1.5	137	0.51	768	8192	62.28
59	text-embedding-3-small			1536	8191	62.26
60	e5-large-v2	335	1.25	1024	512	62.25
61	bge-small-en-v1.5	33	0.12	384	512	62.17
62	Cohere-embed-english-light-v3.0					62.01
63	text-embedding-3-large-256	138	0.51	256	8191	62.00
64	nomi-embed-text-v1.5-512	1280	4.77	512	8192	61.96
65	LLM2Vec-Sheared-Llama-supervised	1241	4.62	2048	4096	61.85
66	instructor-xl	335	1.25	768	512	61.79
67	instructor-large	109	0.41	768	512	61.59
68	e5-base-v2	112	0.42	768	4096	61.50
69	e5-base-4k	560	2.09	1024	514	61.50
70	multilingual-e5-large	335	0.51	768	512	61.42
71	e5-large	137	0.12	384	8192	61.36
72	nomi-embed-text-v1-ablated	33	0.51	256	8191	60.99
73	gte-small	138	0.51	1536	2048	60.63
74	nomi-embed-text-v1.5-256	7069	26.33	4096	512	60.44
75	text-embedding-ada-002	109	0.41	768	8192	60.38
76	udever-bloom-7b1	137	0.51	768	8192	60.37
77	e5-base					60.09
78	jina-embeddings-v2-base-en					60.08
79	Titan-text-embeddings-v2					59.93
80	snowflake-arctic-embed-m-long					
81	Cohere-embed-multilingual-light-v3.0					
82	e5-small-v2					

Continued on next page

Rank	Model	Model Size	Memory Usage	Dimensions	Max Tokens	Avg. Score
83	udaver-bloom-3b	3003	11.19	2560	2048	59.86
84	nomie-embed-text-v1-unsupervised	137	0.51	768	8192	59.85
85	snowflake-arctic-embed-l	334	1.24	1024	512	59.84
86	snowflake-arctic-embed-m	109	0.41	768	512	59.79
87	instructor-base	110	0.41	768	512	59.54
88	sentence-t5-xxl	4865	18.12	768	512	59.51
89	multilingual-e5-base	278	1.04	768	514	59.45
90	nomie-embed-text-v1.5-128	138	0.51	128	8192	59.34
91	XLNet-3B5-embedding					59.29
92	GIST-all-MiniLM-L6-v2	23	0.08	384	512	59.00
93	gtr-t5-xxl	4865	18.12	768	512	58.97
94	snowflake-arctic-embed-s	33	0.12	384	512	58.94
95	SGPT-5.8B-weightedmean-msmarco-specb-bitfit	5874	21.88	4096	2048	58.93
96	e5-small	33	0.12	384	512	58.89
97	gte-tiny	23	0.08	384	512	58.69
98	gtr-t5-xl	1240	4.62	768	512	58.42
99	udaver-bloom-1b1			1536	2048	58.29
100	gtr-t5-large	168	0.63	768	512	58.28

Table 4: Top-100 models on the MTEB leaderboard as of 03.08.2024 at 12:00pm which do not exceed 1 billion parameters. Model size is measured in million parameters.

Rank	Model	Model Size	Is Opensource?	Supports MRL?	Additional Notes
6	stella_en_400M_v5	435	y	y	2D-MRL Model
21	gte-large-en-v1.5	434	y	n	
25	mxbai-embed-large-v1	335	y	y	
26	UAE-Large-V1	335	y	n	
30	multilingual-e5-large-instruct	560	y	n	
32	GIST-large-Embedding-v0	335	y	n	
33	bge-large-en-v1.5	335	y	n	
34	blade-embed	335	y	n	
35	MUG-B-1.6	335	y	n	
37	gte-base-en-v1.5	137	y	n	
38	Cohere-embed-multilingual-v3.0		n	n	
40	bge-base-en-v1.5	438	y	n	
41	privacy_embedding_rag_10k_base_final	109	y	n	
42	privacy_embedding_rag_10k_base_12_final	109	y	n	
43	privacy_embedding_rag_10k_base_15_final	109	y	n	
44	privacy_embedding_rag_10k_base_checkpoint_2	109	y	n	
45	ember-v1	335	y	n	
46	sf_model_e5	335	y	n	
47	mxbai-embed-2d-large-v1	335	y	y	
48	gte-large	335	y	n	
49	NoInstruct-small-Embedding-v0	33	y	n	
50	GIST-small-Embedding-v0	33	y	n	
51	stella-base-en-v2	55	y	n	
52	gte-base	109	y	n	
53	UniVaR-lambda-80	137	y	n	
54	UniVaR-lambda-20	137	y	n	

Continued on next page

Rank	Model	Model Size	Is Opensource?	Supports MRL?	Additional Notes
55	UniVaR-lambda-1	137	y	n	Same model as 58.
56	nomi-embed-text-v1	137	y	n	
57	UniVaR-lambda-5	137	y	n	
58	nomi-embed-text-v1.5	137	y	y	
60	e5-large-v2	335	y	n	
61	bge-small-en-v1.5	33	y	n	
64	nomi-embed-text-v1.5-512	138	y	y	
67	instructor-large	335	y	n	
68	e5-base-v2	109	y	n	
69	e5-base-4k	112	y	n	
70	multilingual-e5-large	560	y	n	Same model as 58.
71	e5-large	335	y	n	
72	nomi-embed-text-v1-ablated	137	y	n	
73	gte-small	33	y	n	
74	nomi-embed-text-v1.5-256	138	y	y	
77	e5-base	109	y	n	
78	jina-embeddings-v2-base-en	137	y	n	
80	snowflake-arctic-embed-m-long	137	y	n	
82	e5-small-v2	33	y	n	
84	nomi-embed-text-v1-unsupervised	137	y	n	Same model as 58.
85	snowflake-arctic-embed-l	334	y	n	
86	snowflake-arctic-embed-m	109	y	n	
87	instructor-base	110	y	n	
89	multilingual-e5-base	278	y	n	
90	nomi-embed-text-v1.5-128	138	y	y	
92	GIST-all-MiniLM-L6-v2	23	y	n	
94	snowflake-arctic-embed-s	33	y	n	
96	e5-small	33	y	n	
97	gte-tiny	23	y	n	

Continued on next page

Rank	Model	Model Size	Is Opensource?	Supports MRL?	Additional Notes
100	gtr-t5-large	168	y	n	

B. Additional Figures from Literature

B.1 MRL Figures

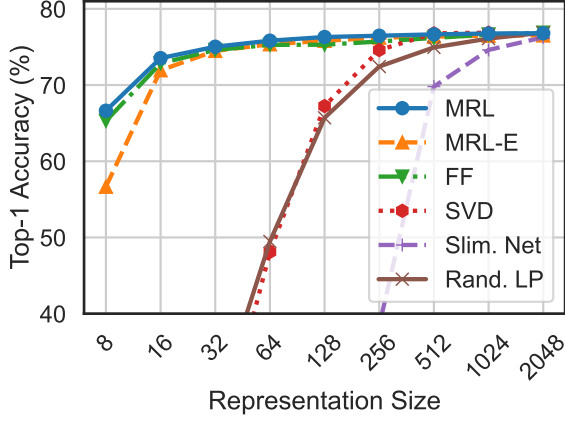


Figure 1: ImageNet-1K linear classification accuracy of ResNet50 models. MRL is as accurate as the independently trained FF models for every representation size. Figure and caption excerpted from the original paper [KBR⁺22].

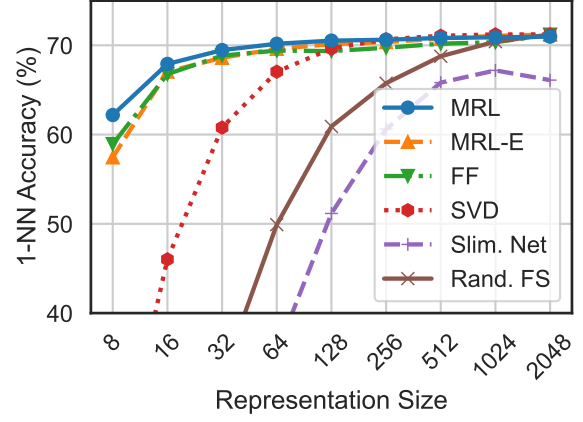


Figure 2: ImageNet-1K 1-NN accuracy of ResNet50 models measuring the representation quality for downstream task. MRL outperforms all the baselines across all representation sizes. Figure and caption excerpted from the original paper [KBR⁺22].

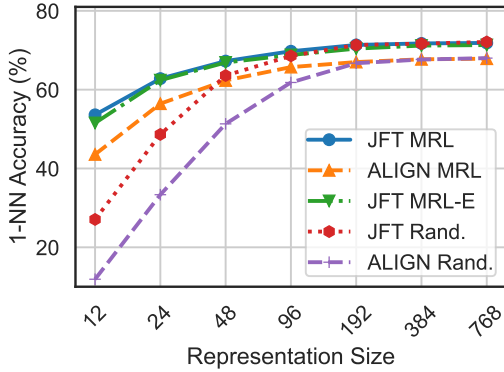


Figure 3: ImageNet-1K 1-NN accuracy for ViT-B/16 models trained on JFT-300M & as part of ALIGN. MRL scales seamlessly to web-scale with minimal training overhead. Figure and caption excerpted from the original paper [KBR⁺22].

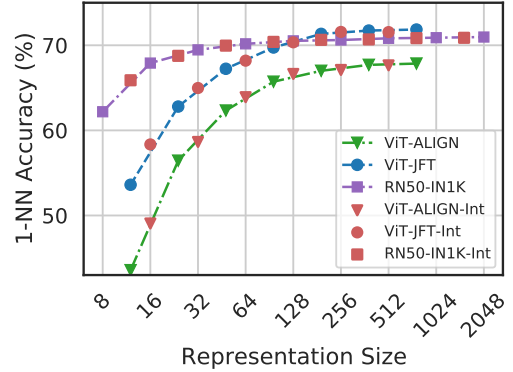


Figure 4: Despite optimizing MRL only for $O(\log(d))$ dimensions for ResNet50 and ViT-B/16 models; the accuracy in the intermediate dimensions shows interpolating behavior. Figure and caption excerpted from the original paper [KBR⁺22].

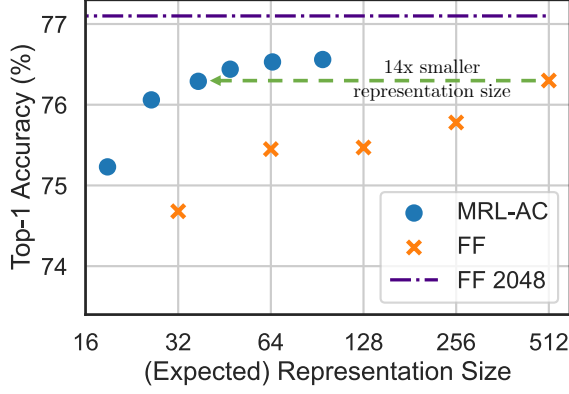


Figure 5: Adaptive classification on MRL ResNet50 using cascades results in $14\times$ smaller representation size for the same level of accuracy on ImageNet-1K (~ 37 vs 512 dims for 76.3%). Figure and caption excerpted from the original paper [KBR⁺22].

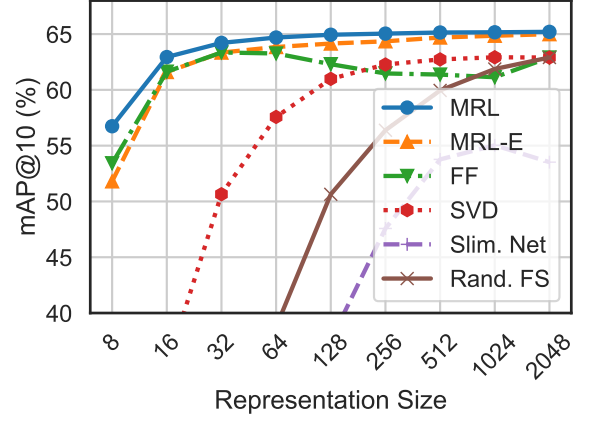
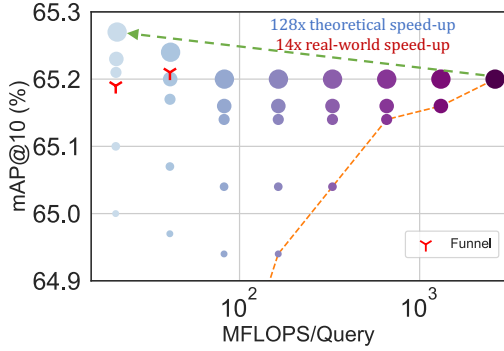
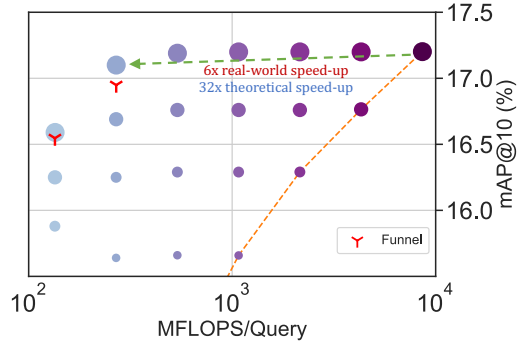


Figure 6: mAP@10 for Image Retrieval on ImageNet-1K with ResNet50. MRL consistently produces better retrieval performance over the baselines across all the representation sizes. Figure and caption excerpted from the original paper [KBR⁺22].



(a) ImageNet-1K



(b) ImageNet-4K

Figure 7: The trade-off between mAP@10 vs MFLOPs/Query for Adaptive Retrieval (AR) on ImageNet-1K (left) and ImageNet-4K (right). Every combination of D_s & D_r falls above the Pareto line (orange dots) of single-shot retrieval with a fixed representation size while having configurations that are as accurate while being up to $14\times$ faster in real-world deployment. Funnel retrieval is almost as accurate as the baseline while alleviating some of the parameter choices of Adaptive Retrieval. Figure and caption excerpted from the original paper [KBR⁺22].

B.2 2DMSE Figures

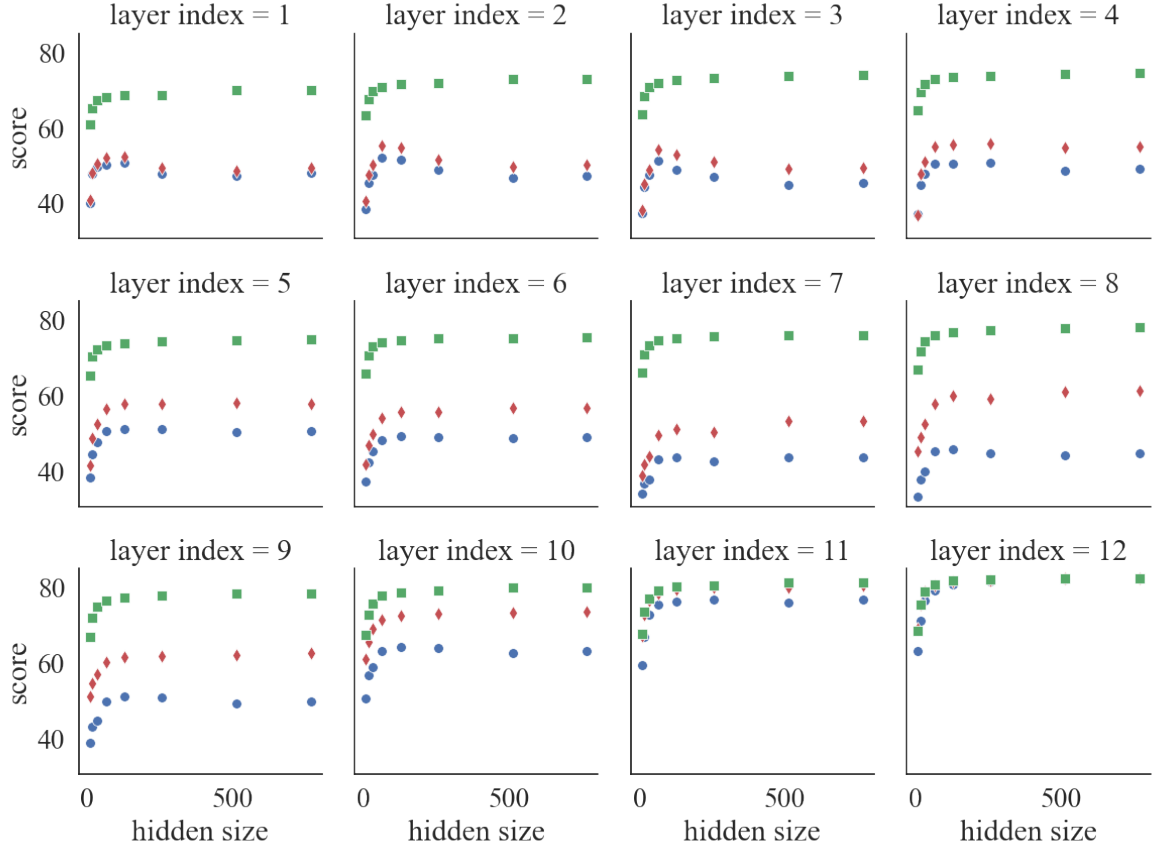
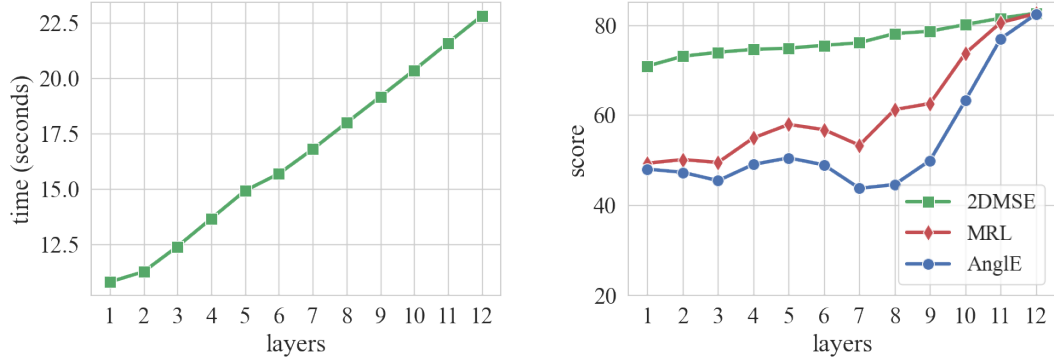


Figure 8: Results of the STS benchmark with a cascade of hidden sizes: $8 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512 \rightarrow 768$ from BERT_{base} . The score represents the average Spearman’s correlation. BERT_{base} serves as the backbone for all models. The blue $\bullet$ indicates the results of sentence embeddings from AngleE without any scalable sentence embedding learning. The red $\blacklozenge$ represents the results of matryoshka sentence embeddings. The green $\blacksquare$ denotes the results of our proposed 2D Matryoshka Sentence Embeddings (2DMSE). The layer index = i denotes the i -th attention layer. Figure and caption excerpted from the original paper [LLL⁺24].



(a) Inference time *vs* number of layers. (b) Score on STS *vs* number of layers.

Figure 9: Subfigure (a) illustrates the time taken to use embeddings from different layers to encode the entire STS benchmarks. Subfigure (b) displays the average Spearman’s correlation scores of different layers. Both (a) and (b) use an embedding size of 768 and the standard STS benchmark dataset. Figure and caption excerpted from the original paper [LLL⁺24].

Model	STS12	STS13	STS14	STS15	STS16	STS-B	SICK-R	Avg.
GloVe [RG19]	52.86	66.75	62.15	72.77	66.87	68.03	65.65	65.01
USE [RG19]	64.49	67.80	64.61	76.83	73.18	74.92	76.69	71.22
SBERT [RG19]	70.97	76.53	73.19	79.09	74.30	77.03	72.91	74.89
SimCSE [GYC21]	75.30	84.67	80.19	85.40	80.82	84.25	80.39	81.57
AngIE [LL23]	75.09	85.56	80.66	86.44	82.47	85.16	81.23	82.37
MRL ($d = 768$) $\star$	75.72	86.79	81.89	86.91	81.74	85.50	79.44	82.57
2DMSE ($n = 12$, $d = 768$)	75.00	86.69	82.30	86.50	82.09	85.79	80.18	82.65

Table 5: Full-capacity sentence embedding performance on the standard STS benchmark. Results $\star$ denote our implementation. BERT_{base} serves as the backbone for all models. Figure and caption excerpted from the original paper [LLL⁺24].

Model	Avg. Spearman's Correlation
$n = 12, d = 768$	
2DMSE	82.65
w/o alignment $\mathcal{L}_{align}$	82.57 (−0.08)
w/o last layer $\mathcal{L}_N^D$	81.31 (−1.34)
$n = 8, d = 512$	
2DMSE	78.02
w/o alignment $\mathcal{L}_{align}$	77.94 (−0.08)
w/o last layer $\mathcal{L}_N^D$	76.52 (−1.50)
$n = 6, d = 384$	
2DMSE	75.21
w/o alignment $\mathcal{L}_{align}$	75.08 (−0.13)
w/o last layer $\mathcal{L}_N^D$	74.98 (−0.23)
$n = 4, d = 256$	
2DMSE	73.93
w/o alignment $\mathcal{L}_{align}$	73.69 (−0.24)
w/o last layer $\mathcal{L}_N^D$	73.00 (−0.93)

Table 6: Ablation study results of 2DMSE on the standard STS benchmark using BERT_{base}. n denotes the number of Transformer layers, and d stands for the embedding dimensions. Figure and caption excerpted from the original paper [LLL⁺24].

C. mxbai-embed-large-v1

This section contains additional figures and tables for the MBAI-1D model.

[illegible]

Table 7: Absolute evaluation time for mxpai-embed-large-v1 on all tasks. Table 2 serves as a legend

[illegible]

Table 8: Relative evaluation time for mxbai-embed-large-v1 on all tasks. Table 2 serves as a legend.

q	m	Total Time	Average All Tasks	Average classification	Average clustering	Average sts	Average pairclass	Average retrieval	Average rerank	Average summ
f32	1024	3989.64	104.99	34.12	100.46	19.21	111.65	501.87	213.42	39.10
f32	512	3887.41	102.30	29.38	97.49	19.20	109.95	494.10	211.41	39.10
f32	256	3830.50	100.80	26.19	95.02	19.17	109.96	493.85	209.53	39.04
f32	128	3785.90	99.63	23.03	93.79	19.11	109.52	494.03	207.65	39.06
f32	64	3827.66	100.73	22.67	93.18	19.11	108.98	501.78	216.72	40.91
f32	32	3828.93	100.76	22.28	93.10	19.11	109.37	502.96	216.89	40.97
f32	16	3814.97	100.39	21.75	92.78	19.11	108.88	503.06	215.13	40.96
f32	8	3808.78	100.23	21.63	92.48	19.13	109.01	502.75	214.50	40.83
int	1024	3781.10	99.50	34.82	100.87	19.35	113.24	443.57	198.04	35.82
int	512	3683.75	96.94	29.58	97.29	19.22	110.28	442.99	195.50	35.77
int	256	3623.37	95.35	26.43	94.33	19.18	110.11	443.42	193.48	35.71
int	128	3586.75	94.39	23.13	93.70	19.09	109.65	443.51	192.89	35.68
int	64	3572.91	94.02	22.63	92.82	19.09	109.06	443.40	193.24	35.62
int	32	3573.05	94.03	22.58	92.75	19.07	109.23	443.63	193.29	35.81
int	16	3560.70	93.70	22.23	92.51	19.04	109.06	443.30	191.57	35.62
int	8	3562.39	93.75	22.17	92.53	19.14	109.11	443.71	191.37	35.72
bin	1024	3585.65	94.36	22.97	93.81	19.12	109.80	443.34	192.51	35.66
bin	512	3576.19	94.11	22.64	93.20	19.10	109.52	443.53	192.43	35.64
bin	256	3568.42	93.91	22.33	92.81	19.05	109.28	443.52	192.37	35.71
bin	128	3568.44	93.91	22.16	92.82	19.08	109.06	443.59	192.78	35.81
bin	64	3559.44	93.67	21.98	92.57	19.06	109.16	443.41	191.26	35.77
bin	32	3552.10	93.48	21.93	92.18	19.08	108.91	443.20	190.65	35.79
bin	16	3550.49	93.43	21.86	92.21	19.09	108.62	443.63	189.99	35.82
bin	8	3545.92	93.31	21.73	92.09	19.08	108.58	443.18	190.02	35.03

Table 9: Evaluation time for mxbai-embed-large-v1, averaged per category. Table 2 serves as a legend.

q	m	Average Relative classification	Average Relative clustering	Average Relative sts	Average Relative pairclass	Average Relative retrieval	Average Relative rerank	Average Relative summ	Average Relative all
f32	1024	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
f32	512	0.884	0.895	0.996	0.965	0.986	0.994	1.000	0.943
f32	256	0.809	0.831	0.998	0.964	0.985	0.988	0.999	0.910
f32	128	0.761	0.792	0.997	0.959	0.986	0.983	0.999	0.888
f32	64	0.753	0.772	0.994	0.947	1.001	1.017	1.046	0.885
f32	32	0.742	0.772	0.994	0.946	1.004	1.020	1.048	0.883
f32	16	0.729	0.763	1.000	0.946	1.004	1.012	1.048	0.879
f32	8	0.727	0.745	0.995	0.944	1.004	1.010	1.044	0.872
int	1024	1.009	1.017	1.004	1.026	0.886	0.919	0.916	0.992
int	512	0.885	0.901	0.998	0.971	0.885	0.911	0.915	0.929
int	256	0.819	0.826	1.002	0.967	0.886	0.903	0.913	0.895
int	128	0.762	0.793	0.996	0.958	0.886	0.902	0.913	0.872
int	64	0.756	0.766	0.996	0.948	0.886	0.903	0.911	0.863
int	32	0.754	0.763	0.994	0.953	0.886	0.903	0.916	0.862
int	16	0.743	0.756	0.994	0.944	0.886	0.897	0.911	0.856
int	8	0.742	0.756	0.995	0.950	0.886	0.896	0.914	0.857
bin	1024	0.756	0.802	0.995	0.963	0.886	0.900	0.912	0.873
bin	512	0.753	0.773	0.994	0.957	0.886	0.900	0.912	0.864
bin	256	0.745	0.763	0.992	0.954	0.886	0.899	0.913	0.859
bin	128	0.738	0.764	0.994	0.949	0.886	0.901	0.916	0.858
bin	64	0.733	0.747	0.995	0.951	0.886	0.896	0.915	0.853
bin	32	0.730	0.740	0.996	0.942	0.885	0.894	0.915	0.850
bin	16	0.727	0.735	0.999	0.943	0.886	0.891	0.916	0.848
bin	8	0.726	0.730	0.994	0.938	0.885	0.891	0.896	0.844

Table 10: Relative evaluation time for mxbai-embed-large-v1, averaged per category. Table 2 serves as a legend.

q	m	req. bits	Average classification	Average clustering	Average sts	Average pairclass	Average retrieval	Average rerank	Average summ	Average all
f32	1024	32768	0.751	0.477	0.847	0.872	0.482	0.510	0.327	0.662
f32	512	16384	0.741	0.475	0.845	0.870	0.476	0.503	0.316	0.658
f32	256	8192	0.728	0.461	0.842	0.866	0.447	0.500	0.308	0.647
f32	128	4096	0.703	0.443	0.834	0.859	0.404	0.492	0.301	0.630
f32	64	2048	0.658	0.404	0.814	0.842	0.329	0.471	0.302	0.596
f32	32	1024	0.590	0.364	0.785	0.811	0.217	0.450	0.285	0.551
f32	16	512	0.487	0.313	0.749	0.743	0.098	0.415	0.278	0.488
f32	8	256	0.384	0.273	0.666	0.604	0.029	0.375	0.265	0.415
int	1024	8192	0.718	0.478	0.841	0.872	0.480	0.503	0.321	0.653
int	512	4096	0.706	0.472	0.839	0.871	0.471	0.498	0.313	0.647
int	256	2048	0.692	0.460	0.836	0.866	0.446	0.496	0.301	0.637
int	128	1024	0.657	0.441	0.829	0.860	0.399	0.491	0.292	0.618
int	64	512	0.591	0.401	0.808	0.843	0.323	0.469	0.295	0.579
int	32	256	0.497	0.361	0.775	0.813	0.207	0.445	0.286	0.527
int	16	128	0.375	0.308	0.728	0.741	0.092	0.405	0.263	0.456
int	8	64	0.328	0.272	0.640	0.614	0.027	0.363	0.255	0.395
bin	1024	1024	0.595	0.378	0.788	0.832	0.245	0.448	0.311	0.561
bin	512	512	0.491	0.315	0.748	0.794	0.139	0.415	0.293	0.497
bin	256	256	0.380	0.281	0.682	0.722	0.062	0.387	0.282	0.433
bin	128	128	0.305	0.258	0.567	0.594	0.027	0.354	0.288	0.366
bin	64	64	0.259	0.248	0.439	0.452	0.019	0.324	0.314	0.306
bin	32	32	0.210	0.245	0.326	0.366	0.011	0.310	0.281	0.256
bin	16	16	0.179	0.244	0.284	0.315	0.011	0.285	0.262	0.232
bin	8	8	0.162	0.241	0.167	0.275	0.001	0.149	0.195	0.180

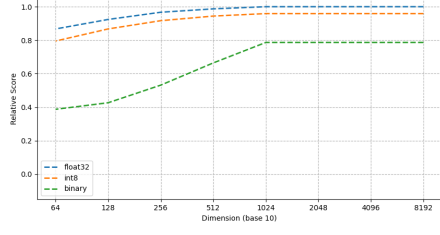
Table 13: Performance for mxbai-embed-large-v1, averaged per category. [Table 2](#) serves as a legend.

q	m	req. bits	Average Relative classification	Average Relative clustering	Average Relative sts	Average Relative pairclass	Average Relative retrieval	Average Relative rerank	Average Relative summ	Average Relative all
f32	1024	32768	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
f32	512	16384	0.987	0.998	0.998	0.998	0.986	0.989	0.968	0.993
f32	256	8192	0.968	0.967	0.995	0.993	0.923	0.978	0.943	0.974
f32	128	4096	0.933	0.931	0.985	0.985	0.827	0.960	0.919	0.944
f32	64	2048	0.871	0.850	0.962	0.965	0.669	0.914	0.924	0.886
f32	32	1024	0.783	0.770	0.928	0.929	0.442	0.871	0.871	0.812
f32	16	512	0.652	0.662	0.886	0.851	0.199	0.812	0.850	0.714
f32	8	256	0.525	0.574	0.790	0.699	0.060	0.738	0.809	0.609
int	1024	8192	0.955	1.002	0.994	1.000	0.996	0.986	0.982	0.988
int	512	4096	0.938	0.991	0.991	0.998	0.975	0.978	0.958	0.977
int	256	2048	0.920	0.965	0.988	0.993	0.921	0.972	0.921	0.960
int	128	1024	0.872	0.928	0.980	0.985	0.820	0.960	0.893	0.927
int	64	512	0.782	0.844	0.954	0.966	0.660	0.916	0.904	0.862
int	32	256	0.661	0.763	0.916	0.931	0.422	0.869	0.875	0.779
int	16	128	0.510	0.652	0.861	0.848	0.182	0.796	0.805	0.671
int	8	64	0.453	0.570	0.759	0.710	0.056	0.718	0.780	0.583
bin	1024	1024	0.786	0.793	0.931	0.953	0.487	0.876	0.952	0.827
bin	512	512	0.654	0.664	0.884	0.909	0.274	0.814	0.897	0.726
bin	256	256	0.512	0.592	0.806	0.826	0.125	0.760	0.863	0.634
bin	128	128	0.424	0.540	0.672	0.685	0.056	0.704	0.882	0.545
bin	64	64	0.365	0.519	0.521	0.527	0.042	0.648	0.960	0.472
bin	32	32	0.302	0.512	0.387	0.430	0.025	0.623	0.861	0.408
bin	16	16	0.261	0.509	0.336	0.373	0.025	0.579	0.800	0.376
bin	8	8	0.240	0.501	0.200	0.328	0.002	0.287	0.597	0.299

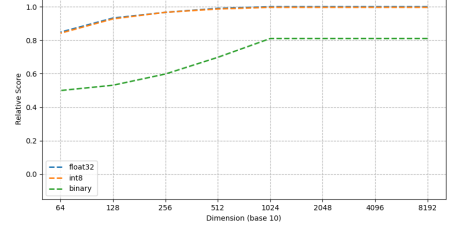
Table 14: Relative performance for mxbai-embed-large-v1, averaged per category. [Table 2](#) serves as a legend.

D. stella_en_400M_v5

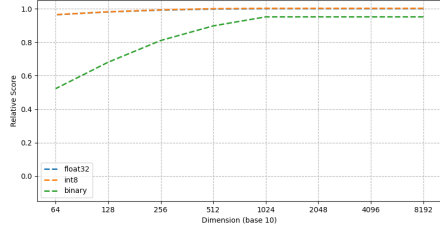
This section contains additional figures and tables for the STELLA model.



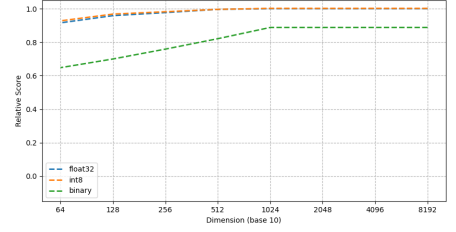
(a) Classification



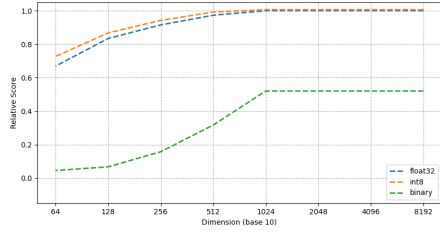
(b) Clustering



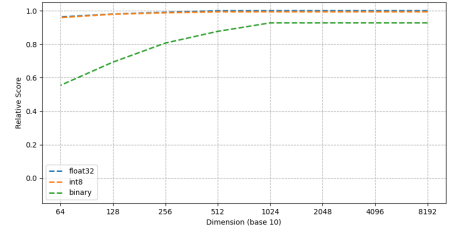
(c) Pair Classification



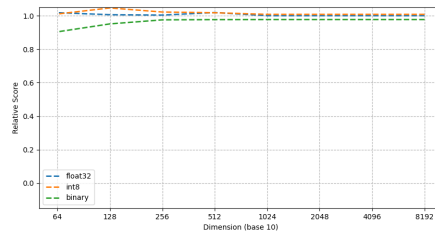
(d) Re-Ranking



(e) Retrieval

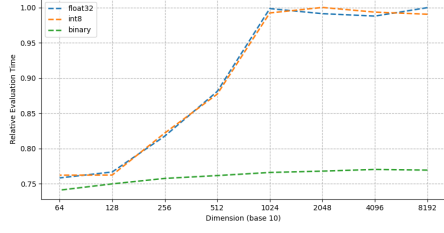


(f) Semantic Textual Similarity

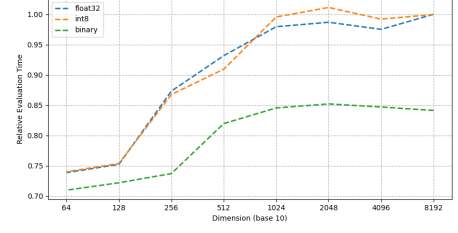


(g) Summarization

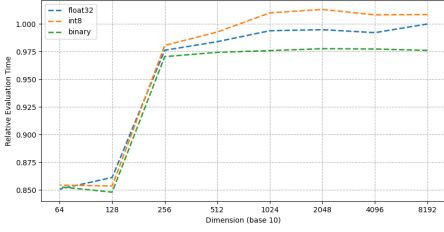
Figure 10: Relative performance of the STELLA model for each task category in the MTEB. The x-axis of each subfigure represents the embedding size while the y-axis represents the relative performance. For each task the reference point is the performance of the float32 embeddings at the full embedding size. Blue: float32, Orange: int8, Green: binary.



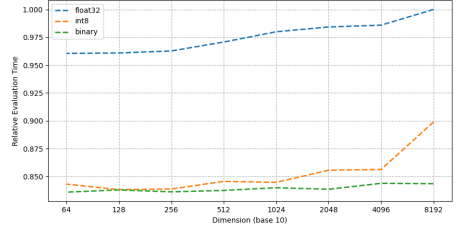
(a) Classification



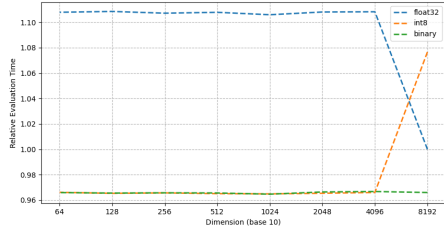
(b) Clustering



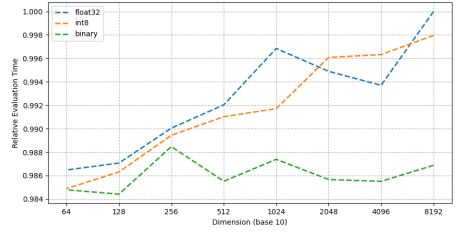
(c) Pair Classification



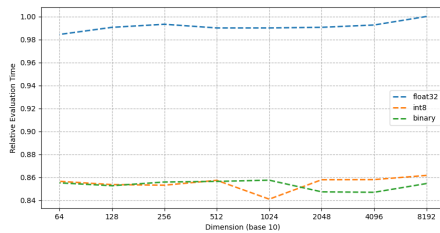
(d) Re-Ranking



(e) Retrieval

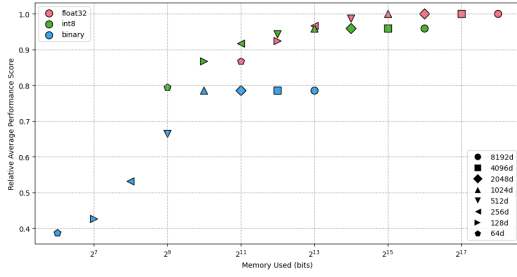


(f) Semantic Textual Similarity

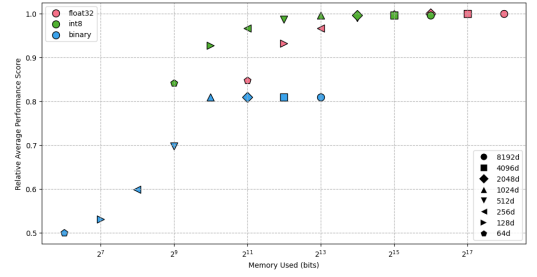


(g) Summarization

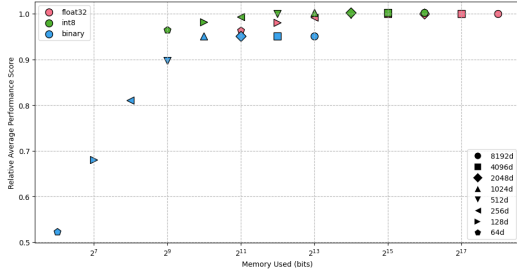
Figure 11: Relative evaluation time of the STELLA model for each task category in the MTEB. The x-axis of each subfigure represents the embedding size while the y-axis represents the relative evaluation time. For each task the reference point is the evaluation time of the float32 embeddings at the full embedding size. Blue: float32, Orange: int8, Green: binary.



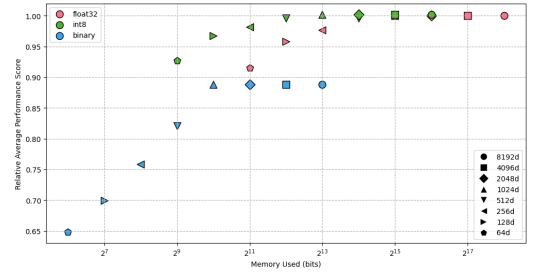
(a) Classification



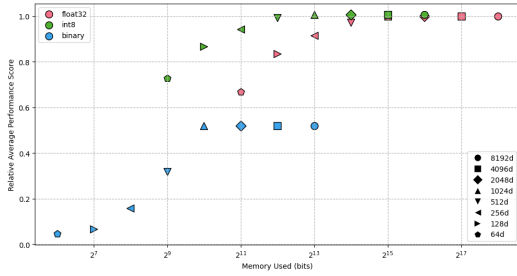
(b) Clustering



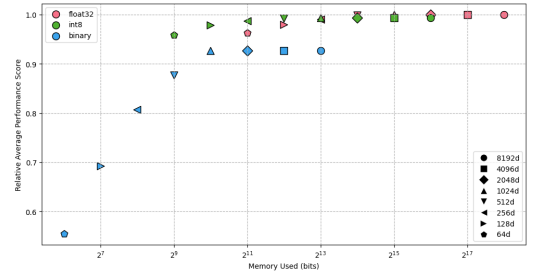
(c) Pair Classification



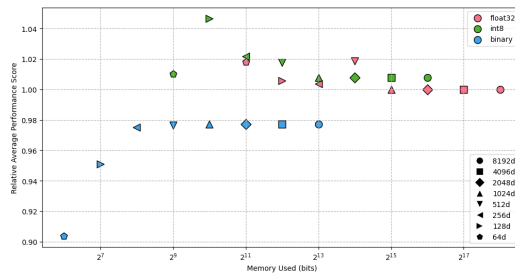
(d) Re-Ranking



(e) Retrieval



(f) Semantic Textual Similarity



(g) Summarization

Figure 12: Relative accuracy compute trade-off of the STELLA model, averaged over all selected MTEB tasks within each category. The x-axis represents memory consumption (log scale) and the y-axis represents relative performance (averaged over tasks within the category). Reference point: float32 embeddings at full size.

q	m	Total Time	Average All Tasks	Average classification	Average clustering	Average sts	Average pairclass	Average retrieval	Average rerank	Average summ
f32	8192	4295.55	113.04	36.26	100.80	22.66	139.56	524.76	241.86	52.36
f32	4096	4463.78	117.47	35.74	101.87	22.57	139.06	582.55	238.88	51.97
f32	2048	4470.14	117.64	35.79	102.71	22.56	138.97	582.55	238.20	51.87
f32	1024	4461.36	117.40	36.29	102.21	22.60	138.89	581.40	236.73	51.84
f32	512	4393.43	115.62	31.13	100.51	22.52	138.27	582.46	233.35	51.84
f32	256	4342.03	114.26	28.06	98.83	22.51	137.80	582.05	230.84	52.01
f32	128	4156.98	109.39	24.78	88.11	22.44	121.08	582.73	229.96	51.86
f32	64	4143.28	109.03	24.33	87.41	22.46	120.48	582.54	229.79	51.54
int	8192	4325.80	113.84	36.46	101.71	22.70	140.11	565.73	209.12	45.12
int	4096	4143.34	109.04	36.43	101.50	22.69	140.22	507.54	207.28	44.93
int	2048	4151.41	109.25	36.85	102.07	22.65	140.19	507.15	207.49	44.92
int	1024	4130.42	108.70	36.44	101.67	22.59	140.12	506.92	203.72	44.04
int	512	4067.18	107.03	31.19	99.73	22.50	139.09	507.07	203.96	44.89
int	256	4019.45	105.78	28.21	98.30	22.51	138.08	507.40	201.51	44.67
int	128	3831.03	100.82	24.67	87.61	22.48	120.55	507.20	201.58	44.70
int	64	3833.36	100.88	24.43	87.48	22.40	120.76	507.51	203.14	44.85
bin	8192	3990.36	105.01	24.93	97.78	22.45	137.78	507.48	202.69	44.75
bin	4096	3995.47	105.14	24.83	98.04	22.44	138.15	507.88	203.22	44.35
bin	2048	3988.21	104.95	24.81	97.97	22.46	137.84	507.61	201.54	44.37
bin	1024	3985.40	104.88	24.76	97.82	22.45	137.74	506.75	202.09	44.90
bin	512	3976.34	104.64	24.40	97.45	22.43	137.44	507.31	201.09	44.85
bin	256	3888.11	102.32	24.19	88.89	22.43	137.20	507.38	200.94	44.82
bin	128	3813.89	100.37	23.91	86.70	22.42	120.31	507.25	201.36	44.65
bin	64	3811.24	100.30	23.73	86.58	22.41	120.63	507.42	200.83	44.78

Table 17: Evaluation time for stella_en_400M_v5, averaged per category. Table 2 serves as a legend.

q	m	Average Relative classification	Average Relative clustering	Average Relative sts	Average Relative pairclass	Average Relative retrieval	Average Relative rerank	Average Relative summ	Average Relative all
f32	8192	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
f32	4096	0.988	0.975	0.994	0.992	1.108	0.986	0.993	0.996
f32	2048	0.992	0.987	0.995	0.995	1.108	0.984	0.991	1.000
f32	1024	0.999	0.980	0.997	0.994	1.106	0.980	0.990	1.000
f32	512	0.881	0.932	0.992	0.984	1.108	0.971	0.990	0.960
f32	256	0.818	0.873	0.990	0.976	1.107	0.963	0.993	0.929
f32	128	0.767	0.752	0.987	0.861	1.109	0.961	0.991	0.877
f32	64	0.758	0.739	0.987	0.851	1.108	0.961	0.984	0.870
int	8192	0.991	1.000	0.998	1.009	1.077	0.899	0.862	0.993
int	4096	0.994	0.992	0.996	1.008	0.966	0.856	0.858	0.979
int	2048	1.000	1.011	0.996	1.013	0.965	0.856	0.858	0.985
int	1024	0.993	0.996	0.992	1.010	0.965	0.845	0.841	0.977
int	512	0.877	0.909	0.991	0.993	0.965	0.846	0.857	0.929
int	256	0.822	0.868	0.989	0.981	0.966	0.839	0.853	0.904
int	128	0.762	0.754	0.986	0.854	0.965	0.838	0.854	0.851
int	64	0.762	0.740	0.985	0.854	0.966	0.843	0.857	0.847
bin	8192	0.769	0.841	0.987	0.976	0.966	0.844	0.855	0.885
bin	4096	0.770	0.847	0.986	0.977	0.967	0.844	0.847	0.887
bin	2048	0.768	0.852	0.986	0.978	0.966	0.839	0.847	0.887
bin	1024	0.766	0.845	0.987	0.976	0.965	0.840	0.858	0.886
bin	512	0.762	0.820	0.986	0.974	0.966	0.838	0.857	0.877
bin	256	0.758	0.737	0.989	0.971	0.966	0.836	0.856	0.855
bin	128	0.750	0.722	0.984	0.848	0.966	0.838	0.853	0.839
bin	64	0.741	0.710	0.985	0.853	0.966	0.836	0.855	0.834

Table 18: Relative evaluation time for stella_en_400M_v5, averaged per category. Table 2 serves as a legend.

q	m	req. bits	Average classification	Average clustering	Average sts	Average pairclass	Average retrieval	Average rerank	Average summ	Average all
f32	8192	262144	0.778	0.500	0.826	0.859	0.477	0.485	0.306	0.664
f32	4096	131072	0.778	0.500	0.826	0.859	0.477	0.485	0.306	0.664
f32	2048	65536	0.778	0.500	0.826	0.859	0.477	0.485	0.306	0.664
f32	1024	32768	0.778	0.500	0.826	0.859	0.477	0.485	0.306	0.664
f32	512	16384	0.769	0.495	0.826	0.857	0.465	0.483	0.312	0.660
f32	256	8192	0.755	0.482	0.818	0.852	0.439	0.475	0.307	0.649
f32	128	4096	0.724	0.465	0.809	0.843	0.401	0.465	0.308	0.631
f32	64	2048	0.680	0.422	0.795	0.828	0.325	0.448	0.311	0.598
int	8192	65536	0.747	0.498	0.821	0.861	0.481	0.486	0.308	0.657
int	4096	32768	0.747	0.498	0.821	0.861	0.481	0.486	0.308	0.657
int	2048	16384	0.747	0.498	0.821	0.861	0.481	0.486	0.308	0.657
int	1024	8192	0.747	0.498	0.821	0.861	0.481	0.486	0.308	0.657
int	512	4096	0.736	0.493	0.821	0.859	0.474	0.482	0.311	0.652
int	256	2048	0.717	0.482	0.816	0.853	0.452	0.476	0.313	0.641
int	128	1024	0.681	0.462	0.809	0.843	0.417	0.469	0.320	0.623
int	64	512	0.625	0.418	0.792	0.829	0.350	0.451	0.309	0.587
bin	8192	8192	0.620	0.403	0.766	0.818	0.254	0.430	0.299	0.565
bin	4096	4096	0.620	0.403	0.766	0.818	0.254	0.430	0.299	0.565
bin	2048	2048	0.620	0.403	0.766	0.818	0.254	0.430	0.299	0.565
bin	1024	1024	0.620	0.403	0.766	0.818	0.254	0.430	0.299	0.565
bin	512	512	0.523	0.347	0.724	0.772	0.157	0.399	0.299	0.505
bin	256	256	0.412	0.298	0.666	0.696	0.078	0.369	0.298	0.438
bin	128	128	0.320	0.266	0.571	0.582	0.033	0.338	0.291	0.371
bin	64	64	0.287	0.251	0.456	0.443	0.022	0.312	0.276	0.315

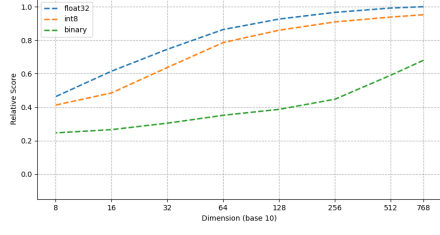
Table 21: Performance for stella_en_400M_v5, averaged per category. Table 2 serves as a legend.

q	m	req. bits	Average Relative classification	Average Relative clustering	Average Relative sts	Average Relative pairclass	Average Relative retrieval	Average Relative rerank	Average Relative summ	Average Relative all
f32	8192	262144	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
f32	4096	131072	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
f32	2048	65536	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
f32	1024	32768	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
f32	512	16384	0.987	0.990	1.000	0.998	0.973	0.996	1.019	0.992
f32	256	8192	0.967	0.966	0.991	0.992	0.915	0.977	1.004	0.973
f32	128	4096	0.924	0.932	0.980	0.981	0.834	0.958	1.006	0.943
f32	64	2048	0.867	0.848	0.963	0.964	0.669	0.916	1.018	0.887
int	8192	65536	0.959	0.996	0.993	1.002	1.006	1.002	1.008	0.990
int	4096	32768	0.959	0.996	0.993	1.002	1.006	1.002	1.008	0.990
int	2048	16384	0.959	0.996	0.993	1.002	1.006	1.002	1.008	0.990
int	1024	8192	0.959	0.996	0.993	1.002	1.006	1.002	1.008	0.990
int	512	4096	0.944	0.987	0.993	1.000	0.992	0.996	1.018	0.982
int	256	2048	0.916	0.967	0.988	0.993	0.942	0.982	1.022	0.964
int	128	1024	0.867	0.927	0.979	0.981	0.867	0.968	1.046	0.934
int	64	512	0.795	0.842	0.959	0.965	0.727	0.927	1.010	0.875
bin	8192	8192	0.786	0.810	0.927	0.951	0.520	0.888	0.977	0.835
bin	4096	4096	0.786	0.810	0.927	0.951	0.520	0.888	0.977	0.835
bin	2048	2048	0.786	0.810	0.927	0.951	0.520	0.888	0.977	0.835
bin	1024	1024	0.786	0.810	0.927	0.951	0.520	0.888	0.977	0.835
bin	512	512	0.664	0.698	0.877	0.897	0.317	0.821	0.977	0.741
bin	256	256	0.532	0.598	0.807	0.810	0.158	0.759	0.975	0.644
bin	128	128	0.426	0.531	0.693	0.681	0.068	0.700	0.951	0.551
bin	64	64	0.387	0.499	0.554	0.523	0.046	0.648	0.904	0.479

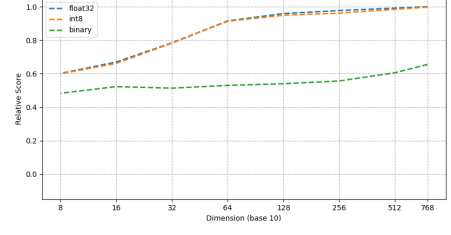
Table 22: Relative performance for stella_en_400M_v5, averaged per category. Table 2 serves as a legend.

E. nomic-embed-text-v1.5

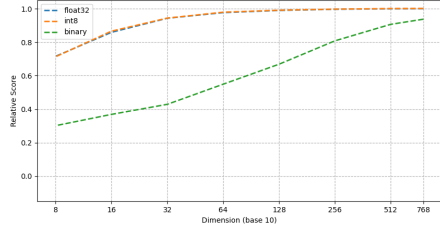
This section contains additional figures and tables for the NOMIC model.



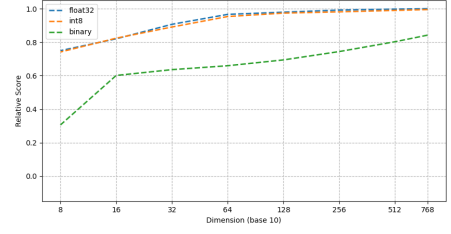
(a) Classification



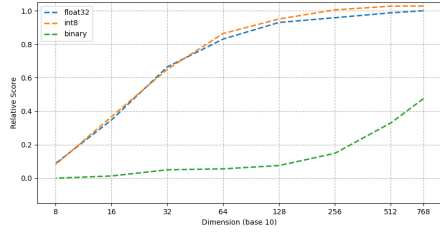
(b) Clustering



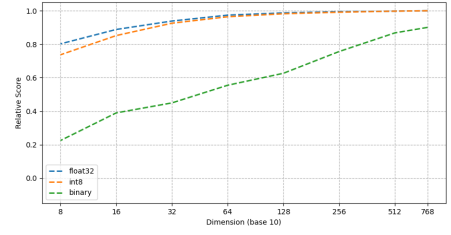
(c) Pair Classification



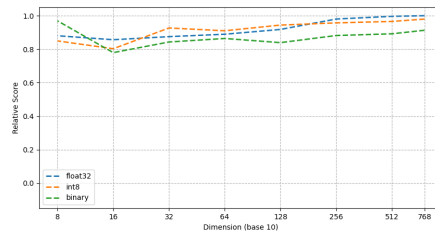
(d) Re-Ranking



(e) Retrieval

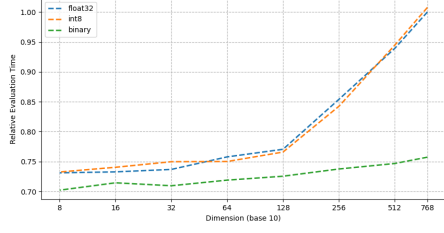


(f) Semantic Textual Similarity

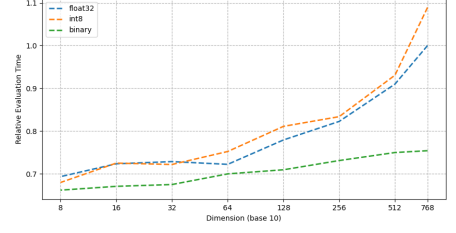


(g) Summarization

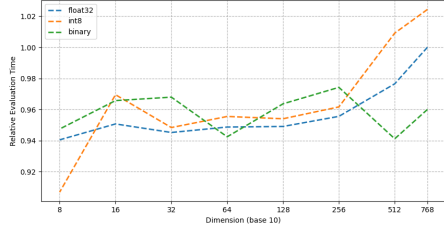
Figure 13: Relative performance of the NOMIC model for each task category in the MTEB. The x-axis of each subfigure represents the embedding size while the y-axis represents the relative performance. For each task the reference point is the performance of the float32 embeddings at the full embedding size. Blue: float32, Orange: int8, Green: binary.



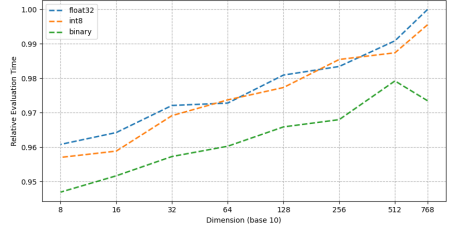
(a) Classification



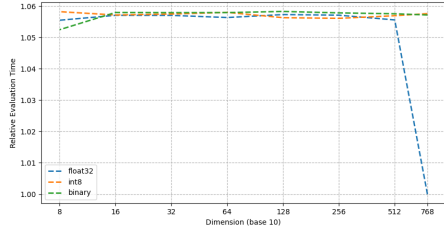
(b) Clustering



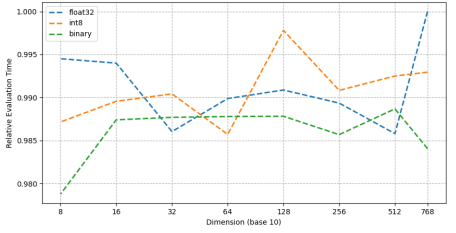
(c) Pair Classification



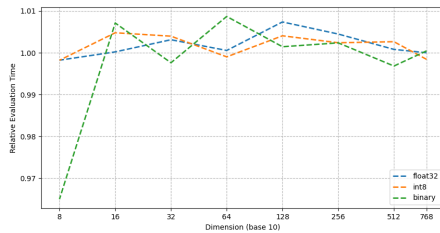
(d) Re-Ranking



(e) Retrieval

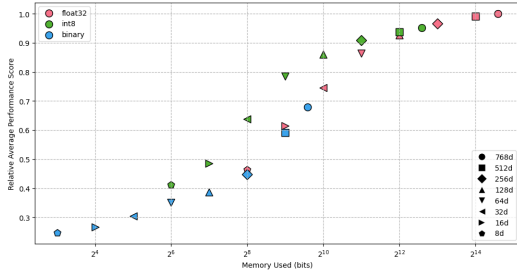


(f) Semantic Textual Similarity

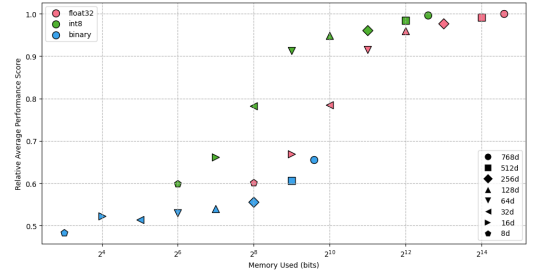


(g) Summarization

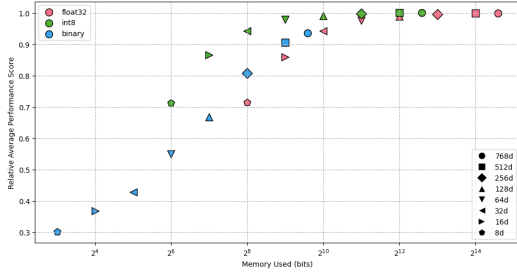
Figure 14: Relative evaluation time of the NOMIC model for each task category in the MTEB. The x-axis of each subfigure represents the embedding size while the y-axis represents the relative evaluation time. For each task the reference point is the evaluation time of the float32 embeddings at the full embedding size. Blue: float32, Orange: int8, Green: binary.



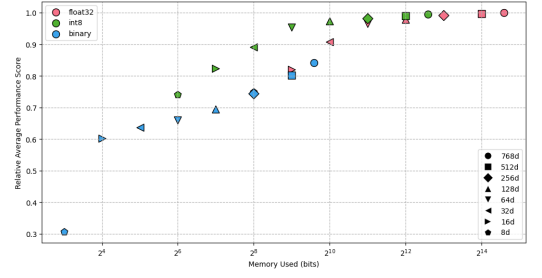
(a) Classification



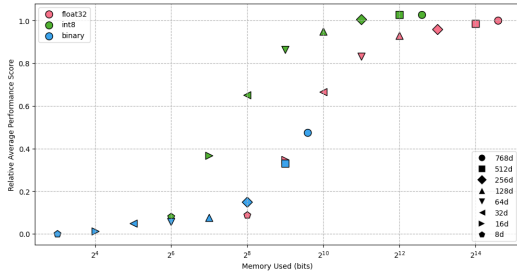
(b) Clustering



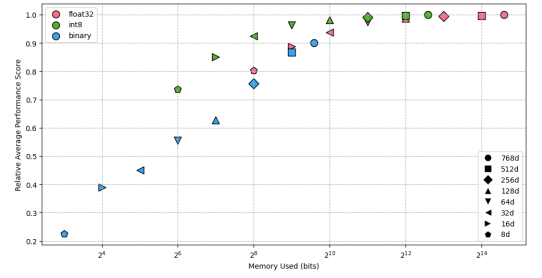
(c) Pair Classification



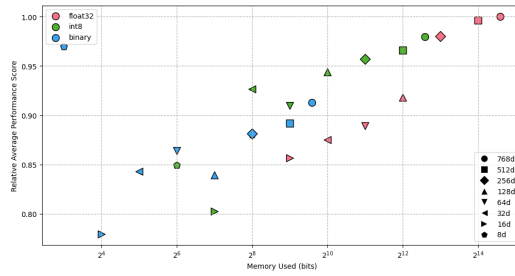
(d) Re-Ranking



(e) Retrieval



(f) Semantic Textual Similarity



(g) Summarization

Figure 15: Relative accuracy compute trade-off of the NOMIC model, averaged over all selected MTEB tasks within each category. The x-axis represents memory consumption (log scale) and the y-axis represents relative performance (averaged over tasks within the category). Reference point: float32 embeddings at full size.

q	m	Total Time	Average All Tasks	Average classification	Average clustering	Average sts	Average pairclass	Average retrieval	Average rerank	Average summ
f32	768	1449.71	42.64	20.00	54.20	8.20	52.10	212.30	113.96	17.58
f32	512	1426.14	41.95	18.27	52.55	8.15	51.36	224.11	112.64	17.59
f32	256	1379.38	40.57	15.28	50.49	8.16	50.78	224.43	111.62	17.66
f32	128	1345.21	39.57	12.10	49.66	8.17	50.75	224.48	111.19	17.71
f32	64	1330.67	39.14	11.64	48.88	8.18	50.90	224.27	109.81	17.59
f32	32	1326.94	39.03	11.37	48.87	8.14	50.58	224.42	109.69	17.63
f32	16	1320.92	38.85	11.25	48.86	8.18	50.62	224.43	107.88	17.58
f32	8	1312.62	38.61	11.18	48.26	8.18	50.38	224.09	107.50	17.55
int	768	1482.13	43.59	20.64	55.71	8.24	52.74	224.54	113.71	17.55
int	512	1425.17	41.92	18.35	52.13	8.23	51.99	224.40	112.39	17.62
int	256	1377.82	40.52	15.12	50.36	8.17	50.93	224.22	111.84	17.62
int	128	1345.79	39.58	12.10	49.92	8.19	50.64	224.26	110.72	17.65
int	64	1332.37	39.19	11.57	49.22	8.16	50.71	224.62	109.70	17.56
int	32	1323.42	38.92	11.43	48.51	8.19	50.67	224.51	109.18	17.65
int	16	1320.31	38.83	11.31	48.79	8.19	50.95	224.45	107.38	17.66
int	8	1299.73	38.23	11.22	47.29	8.15	49.50	224.67	106.75	17.55
bin	768	1333.40	39.22	11.83	49.10	8.16	50.67	224.45	109.76	17.59
bin	512	1330.04	39.12	11.47	48.92	8.17	50.11	224.53	110.71	17.52
bin	256	1326.59	39.02	11.38	48.80	8.18	50.91	224.59	109.26	17.62
bin	128	1319.24	38.80	11.22	48.47	8.20	50.72	224.69	108.32	17.60
bin	64	1312.76	38.61	11.08	48.18	8.17	50.62	224.61	107.58	17.73
bin	32	1309.42	38.51	10.99	48.17	8.19	50.87	224.60	106.49	17.54
bin	16	1306.26	38.42	11.08	47.92	8.20	51.01	224.62	105.72	17.70
bin	8	1294.65	38.08	10.81	47.66	8.11	50.36	223.45	104.90	16.96

Table 25: Evaluation time for nomic-embed-text-v1.5, averaged per category. Table 2 serves as a legend.

q	m	Average Relative classification	Average Relative clustering	Average Relative sts	Average Relative pairclass	Average Relative retrieval	Average Relative rerank	Average Relative summ	Average Relative all
f32	768	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
f32	512	0.939	0.910	0.986	0.977	1.056	0.991	1.001	0.957
f32	256	0.854	0.823	0.989	0.956	1.057	0.983	1.004	0.912
f32	128	0.771	0.779	0.991	0.949	1.057	0.981	1.007	0.881
f32	64	0.758	0.722	0.990	0.949	1.056	0.973	1.001	0.862
f32	32	0.737	0.729	0.986	0.945	1.057	0.972	1.003	0.857
f32	16	0.733	0.723	0.994	0.951	1.057	0.964	1.000	0.857
f32	8	0.731	0.693	0.995	0.941	1.056	0.961	0.998	0.847
int	768	1.008	1.089	0.993	1.024	1.058	0.996	0.998	1.027
int	512	0.945	0.931	0.993	1.009	1.057	0.987	1.003	0.968
int	256	0.842	0.834	0.991	0.962	1.056	0.986	1.002	0.914
int	128	0.766	0.811	0.998	0.954	1.056	0.977	1.004	0.890
int	64	0.750	0.752	0.986	0.956	1.058	0.974	0.999	0.867
int	32	0.750	0.722	0.990	0.948	1.058	0.969	1.004	0.860
int	16	0.741	0.725	0.990	0.970	1.057	0.959	1.005	0.859
int	8	0.733	0.680	0.987	0.907	1.058	0.957	0.998	0.839
bin	768	0.757	0.754	0.984	0.960	1.057	0.974	1.000	0.869
bin	512	0.747	0.750	0.989	0.941	1.058	0.979	0.997	0.866
bin	256	0.738	0.731	0.986	0.974	1.058	0.968	1.002	0.860
bin	128	0.726	0.709	0.988	0.964	1.058	0.966	1.001	0.851
bin	64	0.719	0.700	0.988	0.943	1.058	0.960	1.009	0.845
bin	32	0.710	0.675	0.988	0.968	1.058	0.957	0.998	0.837
bin	16	0.715	0.671	0.987	0.966	1.058	0.952	1.007	0.837
bin	8	0.702	0.662	0.979	0.948	1.052	0.947	0.965	0.826

Table 26: Relative evaluation time for nomic-embed-text-v1.5, averaged per category. Table 2 serves as a legend.

	q	m	req.	CIA	CIA	CIA	CIA	CIA	CIA	CIA	CLU	CLU	CLU	CLU	CLS	STS	STS	STS	STS	STS	PCL	PCL	RET	RER	RER	SUM															
		bits	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	9	10	1	2	3	4	5	6	7	8	9	10	1	2	3	4	5						
f332	768	245	76	0.848	0.488	0.577	0.734	0.720	0.763	0.928	0.733	0.588	0.531	0.413	0.329	Nan	0.327	0.484	0.492	0.394	0.433	0.862	0.785	0.745	0.586	0.804	0.872	0.836	0.843	0.881	Nan	0.956	0.736	0.860	0.536	Nan	0.617	0.310	0.504	0.325	
f332	512	163	84	0.848	0.485	0.573	0.721	0.714	0.758	0.928	0.723	0.588	0.520	0.399	0.328	Nan	0.331	0.484	0.482	0.394	0.437	0.856	0.785	0.743	0.583	0.803	0.871	0.834	0.840	0.878	Nan	0.956	0.736	0.859	0.529	Nan	0.614	0.306	0.504	0.324	
f332	256	81	92	0.843	0.458	0.554	0.710	0.698	0.741	0.908	0.699	0.578	0.518	0.396	0.314	Nan	0.319	0.465	0.482	0.392	0.434	0.847	0.785	0.740	0.584	0.802	0.868	0.830	0.837	0.870	Nan	0.955	0.734	0.856	0.514	Nan	0.614	0.304	0.500	0.319	
f332	128	40	96	0.835	0.428	0.538	0.670	0.644	0.715	0.881	0.657	0.570	0.522	0.383	0.298	Nan	0.318	0.455	0.482	0.387	0.424	0.835	0.783	0.740	0.589	0.795	0.860	0.828	0.832	0.867	Nan	0.947	0.727	0.852	0.498	Nan	0.609	0.303	0.491	0.289	
f332	64	20	104	0.810	0.394	0.504	0.636	0.611	0.666	0.835	0.578	0.508	0.518	0.364	0.273	Nan	0.308	0.412	0.460	0.380	0.387	0.799	0.781	0.737	0.581	0.818	0.824	0.825	0.834	0.863	Nan	0.938	0.713	0.843	0.445	Nan	0.612	0.297	0.479	0.289	
f332	32	10	212	0.736	0.339	0.474	0.599	0.494	0.547	0.712	0.436	0.543	0.513	0.308	0.219	Nan	0.282	0.278	0.364	0.380	0.272	0.762	0.767	0.717	0.787	0.748	0.811	0.785	0.793	0.845	Nan	0.904	0.682	0.823	0.386	Nan	0.584	0.278	0.442	0.282	
f332	16	5	428	0.586	0.277	0.456	0.484	0.345	0.406	0.578	0.314	0.532	0.484	0.238	0.164	Nan	0.255	0.175	0.284	0.374	0.380	0.653	0.737	0.708	0.760	0.711	0.760	0.741	0.769	0.801	Nan	0.777	0.628	0.783	0.283	Nan	0.541	0.251	0.391	0.278	
f332	8	2	864	0.388	0.274	0.436	0.509	0.191	0.260	0.395	0.153	0.505	0.481	0.190	0.138	Nan	0.234	0.127	0.243	0.366	0.161	0.507	0.680	0.677	0.679	0.651	0.699	0.683	0.702	0.705	Nan	0.552	0.543	0.716	0.047	Nan	0.501	0.244	0.329	0.286	
int	768	61	44	0.836	0.463	0.563	0.640	0.691	0.751	0.912	0.674	0.582	0.526	0.403	0.331	Nan	0.328	0.462	0.488	0.392	0.436	0.853	0.782	0.747	0.46	0.835	0.801	0.871	0.828	0.843	0.893	Nan	0.950	0.735	0.861	0.551	Nan	0.613	0.307	0.504	0.318
int	512	256	104	0.836	0.455	0.557	0.616	0.686	0.725	0.916	0.657	0.575	0.519	0.397	0.321	Nan	0.318	0.483																							

Table 27: Absolute performance for nomic-embed-text-v1.5 on all tasks. Table 2 serves as a legend

[illegible]

Table 28: Relative performance for nomic-embed-text-v1.5 on all tasks. Table 2 serves as a legend.

q	m	req. bits	Average classification	Average clustering	Average sts	Average pairclass	Average retrieval	Average rerank	Average summ	Average all
f32	768	24576	0.724	0.444	0.832	0.851	0.536	0.477	0.325	0.650
f32	512	16384	0.718	0.441	0.829	0.850	0.529	0.476	0.324	0.647
f32	256	8192	0.701	0.434	0.827	0.848	0.514	0.474	0.319	0.640
f32	128	4096	0.673	0.427	0.821	0.842	0.498	0.468	0.298	0.628
f32	64	2048	0.629	0.408	0.809	0.831	0.445	0.463	0.289	0.606
f32	32	1024	0.542	0.351	0.780	0.803	0.356	0.435	0.284	0.555
f32	16	512	0.444	0.301	0.737	0.729	0.186	0.394	0.278	0.492
f32	8	256	0.325	0.272	0.665	0.603	0.047	0.358	0.286	0.419
int	768	6144	0.690	0.442	0.831	0.852	0.551	0.475	0.318	0.642
int	512	4096	0.681	0.437	0.829	0.851	0.550	0.473	0.314	0.638
int	256	2048	0.661	0.427	0.824	0.849	0.539	0.468	0.311	0.628
int	128	1024	0.626	0.422	0.816	0.843	0.509	0.465	0.307	0.615
int	64	512	0.574	0.407	0.801	0.833	0.463	0.455	0.296	0.591
int	32	256	0.464	0.350	0.769	0.804	0.348	0.426	0.301	0.533
int	16	128	0.346	0.297	0.707	0.736	0.197	0.393	0.261	0.460
int	8	64	0.286	0.271	0.611	0.601	0.044	0.353	0.276	0.394
bin	768	768	0.496	0.294	0.749	0.799	0.254	0.404	0.297	0.515
bin	512	512	0.429	0.273	0.722	0.773	0.177	0.385	0.290	0.480
bin	256	256	0.320	0.253	0.629	0.689	0.080	0.355	0.286	0.412
bin	128	128	0.270	0.246	0.521	0.565	0.040	0.329	0.273	0.355
bin	64	64	0.244	0.243	0.462	0.460	0.030	0.308	0.281	0.321
bin	32	32	0.208	0.235	0.373	0.355	0.027	0.297	0.274	0.276
bin	16	16	0.177	0.239	0.322	0.303	0.007	0.281	0.253	0.249
bin	8	8	0.163	0.221	0.185	0.245	0.000	0.151	0.315	0.190

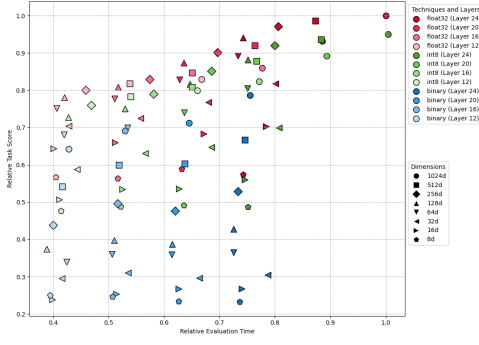
Table 29: Performance for nomic-embed-text-v1.5, averaged per category. [Table 2](#) serves as a legend.

q	m	req. bits	Average Relative classification	Average Relative clustering	Average Relative sts	Average Relative pairclass	Average Relative retrieval	Average Relative rerank	Average Relative summ	Average Relative all
f32	768	24576	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
f32	512	16384	0.992	0.991	0.997	1.000	0.987	0.998	0.996	0.994
f32	256	8192	0.966	0.977	0.994	0.997	0.958	0.992	0.980	0.982
f32	128	4096	0.926	0.959	0.987	0.990	0.930	0.979	0.918	0.961
f32	64	2048	0.864	0.915	0.974	0.977	0.831	0.966	0.889	0.925
f32	32	1024	0.746	0.784	0.938	0.943	0.664	0.907	0.875	0.840
f32	16	512	0.615	0.670	0.888	0.859	0.348	0.820	0.857	0.741
f32	8	256	0.463	0.602	0.802	0.716	0.088	0.750	0.880	0.638
int	768	6144	0.952	0.997	1.000	1.001	1.027	0.995	0.980	0.988
int	512	4096	0.938	0.984	0.997	1.001	1.027	0.990	0.966	0.980
int	256	2048	0.909	0.961	0.991	0.998	1.005	0.981	0.957	0.963
int	128	1024	0.860	0.949	0.982	0.991	0.951	0.974	0.944	0.942
int	64	512	0.786	0.913	0.964	0.979	0.864	0.953	0.910	0.904
int	32	256	0.637	0.782	0.925	0.944	0.650	0.891	0.927	0.810
int	16	128	0.485	0.661	0.851	0.866	0.367	0.824	0.803	0.698
int	8	64	0.412	0.600	0.737	0.713	0.081	0.742	0.850	0.606
bin	768	768	0.680	0.655	0.900	0.937	0.474	0.842	0.913	0.770
bin	512	512	0.591	0.606	0.868	0.907	0.330	0.803	0.892	0.716
bin	256	256	0.448	0.556	0.756	0.808	0.148	0.744	0.882	0.620
bin	128	128	0.387	0.540	0.626	0.669	0.075	0.694	0.839	0.547
bin	64	64	0.352	0.530	0.555	0.549	0.056	0.659	0.864	0.504
bin	32	32	0.304	0.513	0.450	0.429	0.050	0.636	0.843	0.447
bin	16	16	0.266	0.523	0.390	0.369	0.013	0.602	0.779	0.413
bin	8	8	0.247	0.483	0.225	0.301	0.000	0.306	0.969	0.327

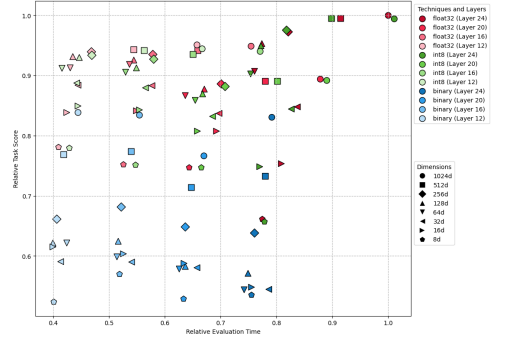
Table 30: Relative performance for nomic-embed-text-v1.5, averaged per category. [Table 2](#) serves as a legend.

F. mxbai-embed-2d-large-v1

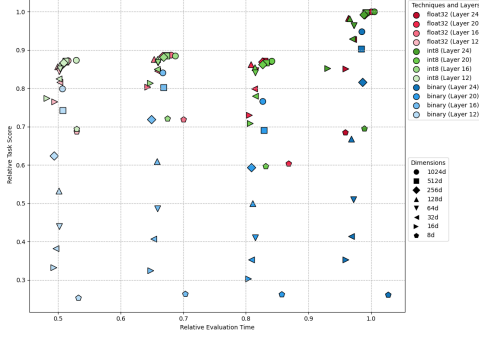
This section contains additional figures and tables for the MBAI-2D model.



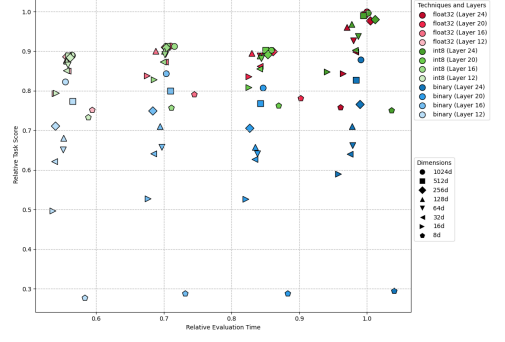
(a) Classification



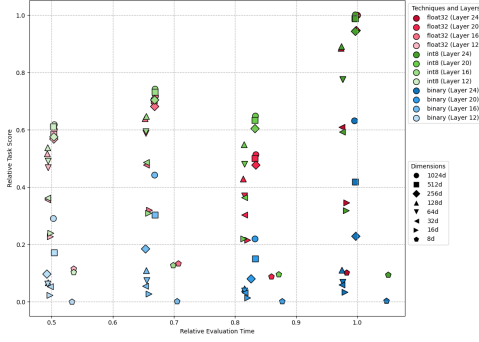
(b) Clustering



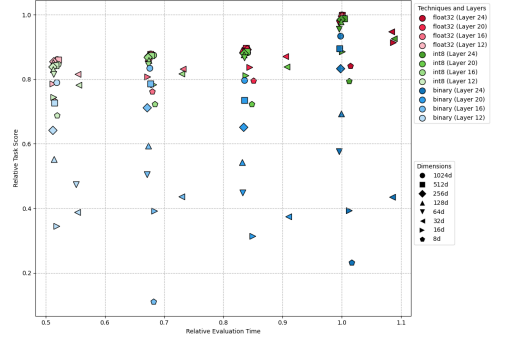
(c) Pair Classification



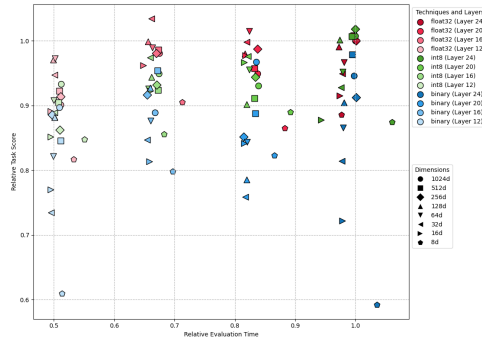
(d) Re-Ranking



(e) Retrieval



(f) Semantic Textual Similarity



(g) Summarization

Figure 16: Trade-off between memory consumption, evaluation time, and performance of MBAI-2D, averaged over all selected MTEB tasks for each task category. The x-axis represents the relative evaluation time while the y-axis represents the relative performance. The color of the markers represent the quantization technique and inference layer, while the shape represents the embedding size of the respective MR-quantization combination.

Table 33: Absolute evaluation time for mx-bai-embed-2d-large-v1 on all tasks using binary quantization. Table 2 serves as a legend

[illegible]

Table 34: Relative evaluation time for mxpai-embed-2d-large-v1 on all tasks using float quantization. Table 2 serves as a legend.

Table 35: Relative evaluation time for mxpai-embed-2d-large-v1 on all tasks using int quantization. Table 2 serves as a legend.

[illegible]

Table 36: Relative evaluation time for mx-bai-embbed-2d-large-v1 on all tasks using binary quantization. Table 2 serves as a legend.

Table 39: Absolute performance for mxbai-embed-2d-large-v1 on all tasks using float quantization. Table 2 serves as a legend

[illegible]

Table 40: Absolute performance for mxbai-embed-2d-large-v1 on all tasks using int quantization. [Table 2](#) serves as a legend.

Table 41: Absolute performance for mxpai-embed-2d-large-v1 on all tasks using binary quantization. [Table 2](#) serves as a legend.

[illegible]

Table 42: Relative performance for mx-bai-embed-2d-large-v1 on all tasks using float quantization. Table 2 serves as a legend.

Table 43: Relative performance for mxbai-embed-2d-large-v1 on all tasks using int quantization. Table 2 serves as a legend

[illegible]

Table 44: Relative performance for mxbai-embed-2d-large-v1 on all tasks using binary quantization. Table 2 serves as a legend

q	m	l	req. bits	Average classification	Average clustering	Average sts	Average pairclass	Average retrieval	Average rerank	Average summ	Average all
f32	1024	24	32768	0.736	0.478	0.845	0.859	0.459	0.496	0.316	0.654
f32	1024	20	32768	0.687	0.430	0.756	0.744	0.244	0.443	0.300	0.578
f32	1024	16	32768	0.636	0.453	0.742	0.768	0.332	0.447	0.310	0.579
f32	1024	12	32768	0.617	0.455	0.727	0.756	0.283	0.431	0.285	0.564
f32	512	24	16384	0.726	0.476	0.843	0.858	0.455	0.492	0.319	0.651
f32	512	20	16384	0.679	0.428	0.756	0.745	0.240	0.444	0.303	0.575
f32	512	16	16384	0.627	0.449	0.740	0.767	0.326	0.445	0.312	0.575
f32	512	12	16384	0.609	0.451	0.725	0.753	0.276	0.429	0.292	0.561
f32	256	24	8192	0.715	0.466	0.841	0.853	0.437	0.485	0.316	0.643
f32	256	20	8192	0.666	0.426	0.754	0.744	0.232	0.443	0.312	0.571
f32	256	16	8192	0.614	0.447	0.736	0.765	0.318	0.445	0.310	0.569
f32	256	12	8192	0.596	0.449	0.721	0.749	0.267	0.430	0.289	0.555
f32	128	24	4096	0.694	0.455	0.835	0.845	0.410	0.478	0.313	0.630
f32	128	20	4096	0.646	0.421	0.752	0.739	0.211	0.440	0.310	0.563
f32	128	16	4096	0.600	0.441	0.730	0.758	0.301	0.440	0.316	0.561
f32	128	12	4096	0.581	0.445	0.719	0.741	0.247	0.426	0.307	0.548
f32	64	24	2048	0.659	0.433	0.821	0.828	0.364	0.463	0.306	0.607
f32	64	20	2048	0.612	0.416	0.747	0.726	0.186	0.436	0.321	0.550
f32	64	16	2048	0.576	0.438	0.721	0.750	0.278	0.436	0.313	0.550
f32	64	12	2048	0.559	0.436	0.709	0.731	0.225	0.421	0.307	0.536
f32	32	24	1024	0.604	0.406	0.800	0.798	0.290	0.448	0.300	0.574
f32	32	20	1024	0.567	0.403	0.734	0.683	0.155	0.422	0.316	0.526
f32	32	16	1024	0.538	0.422	0.701	0.731	0.228	0.426	0.327	0.527
f32	32	12	1024	0.522	0.423	0.688	0.706	0.176	0.412	0.299	0.512
f32	16	24	512	0.516	0.361	0.770	0.732	0.171	0.418	0.289	0.518
f32	16	20	512	0.504	0.389	0.706	0.623	0.112	0.408	0.309	0.493
f32	16	16	512	0.488	0.404	0.680	0.695	0.158	0.407	0.304	0.496
f32	16	12	512	0.475	0.402	0.662	0.660	0.115	0.385	0.282	0.479
f32	8	24	256	0.418	0.319	0.708	0.583	0.053	0.374	0.280	0.445
f32	8	20	256	0.430	0.361	0.671	0.509	0.048	0.381	0.273	0.443
f32	8	16	256	0.416	0.364	0.641	0.620	0.068	0.385	0.286	0.444
f32	8	12	256	0.417	0.376	0.616	0.591	0.059	0.363	0.258	0.436
int	1024	24	8192	0.700	0.476	0.837	0.859	0.458	0.495	0.319	0.644
int	1024	20	8192	0.659	0.428	0.746	0.745	0.300	0.445	0.294	0.573
int	1024	16	8192	0.611	0.449	0.738	0.766	0.342	0.446	0.300	0.572
int	1024	12	8192	0.597	0.452	0.716	0.757	0.288	0.432	0.295	0.557
int	512	24	4096	0.690	0.476	0.835	0.857	0.453	0.491	0.318	0.641
int	512	20	4096	0.649	0.428	0.749	0.741	0.293	0.445	0.288	0.571
int	512	16	4096	0.601	0.447	0.738	0.765	0.337	0.446	0.292	0.568
int	512	12	4096	0.586	0.451	0.713	0.755	0.284	0.431	0.286	0.553
int	256	24	2048	0.678	0.467	0.832	0.853	0.433	0.486	0.322	0.633
int	256	20	2048	0.631	0.424	0.744	0.738	0.281	0.439	0.298	0.563
int	256	16	2048	0.587	0.443	0.733	0.763	0.324	0.444	0.294	0.562
int	256	12	2048	0.569	0.447	0.708	0.751	0.269	0.429	0.273	0.545
int	128	24	1024	0.650	0.454	0.827	0.844	0.410	0.479	0.317	0.619
int	128	20	1024	0.606	0.418	0.740	0.731	0.259	0.436	0.285	0.552
int	128	16	1024	0.559	0.435	0.725	0.758	0.300	0.439	0.298	0.549
int	128	12	1024	0.546	0.445	0.701	0.745	0.255	0.425	0.279	0.536
int	64	24	512	0.598	0.431	0.807	0.828	0.360	0.464	0.301	0.590
int	64	20	512	0.551	0.412	0.732	0.720	0.229	0.432	0.302	0.534
int	64	16	512	0.520	0.431	0.714	0.751	0.276	0.435	0.293	0.534
int	64	12	512	0.510	0.436	0.688	0.737	0.232	0.420	0.287	0.520
int	32	24	256	0.518	0.404	0.783	0.799	0.279	0.446	0.293	0.549
int	32	20	256	0.481	0.400	0.708	0.665	0.176	0.416	0.308	0.500
int	32	16	256	0.468	0.420	0.689	0.735	0.227	0.425	0.308	0.508
int	32	12	256	0.439	0.425	0.659	0.713	0.175	0.411	0.281	0.487
int	16	24	128	0.407	0.359	0.748	0.734	0.153	0.416	0.278	0.487
int	16	20	128	0.393	0.389	0.684	0.602	0.111	0.393	0.305	0.460
int	16	16	128	0.393	0.404	0.661	0.704	0.147	0.402	0.292	0.469
int	16	12	128	0.371	0.407	0.627	0.669	0.117	0.384	0.269	0.450
int	8	24	64	0.351	0.317	0.669	0.591	0.046	0.369	0.276	0.420
int	8	20	64	0.353	0.361	0.610	0.502	0.051	0.370	0.281	0.410
int	8	16	64	0.354	0.363	0.608	0.622	0.065	0.365	0.271	0.420
int	8	12	64	0.346	0.375	0.579	0.598	0.051	0.354	0.268	0.410
bin	1024	24	1024	0.582	0.398	0.788	0.816	0.298	0.439	0.299	0.565
bin	1024	20	1024	0.527	0.368	0.673	0.651	0.109	0.397	0.306	0.484
bin	1024	16	1024	0.514	0.399	0.704	0.727	0.208	0.410	0.281	0.512
bin	1024	12	1024	0.478	0.401	0.665	0.692	0.141	0.395	0.284	0.486
bin	512	24	512	0.491	0.350	0.755	0.777	0.201	0.410	0.309	0.512
bin	512	20	512	0.444	0.343	0.619	0.584	0.074	0.377	0.281	0.436
bin	512	16	512	0.444	0.371	0.663	0.693	0.145	0.389	0.302	0.470
bin	512	12	512	0.402	0.369	0.612	0.641	0.085	0.371	0.267	0.436
bin	256	24	256	0.384	0.304	0.702	0.701	0.110	0.377	0.288	0.447
bin	256	20	256	0.346	0.311	0.549	0.498	0.040	0.347	0.269	0.376
bin	256	16	256	0.363	0.327	0.601	0.620	0.088	0.363	0.290	0.413
bin	256	12	256	0.320	0.318	0.541	0.535	0.047	0.342	0.280	0.374
bin	128	24	128	0.302	0.274	0.584	0.569	0.053	0.346	0.286	0.373
bin	128	20	128	0.272	0.280	0.457	0.416	0.023	0.320	0.248	0.318
bin	128	16	128	0.285	0.299	0.502	0.521	0.051	0.344	0.293	0.351
bin	128	12	128	0.268	0.299	0.463	0.452	0.032	0.325	0.279	0.328
bin	64	24	64	0.255	0.262	0.485	0.429	0.031	0.320	0.274	0.319
bin	64	20	64	0.251	0.277	0.379	0.338	0.020	0.311	0.267	0.285
bin	64	16	64	0.255	0.287	0.426	0.409	0.035	0.317	0.277	0.309
bin	64	12	64	0.240	0.300	0.397	0.371	0.030	0.312	0.260	0.297
bin	32	24	32	0.210	0.263	0.368	0.345	0.027	0.305	0.257	0.271
bin	32	20	32	0.206	0.281	0.317	0.290	0.015	0.298	0.240	0.255
bin	32	16	32	0.217	0.284	0.367	0.341	0.025	0.306	0.268	0.277
bin	32	12	32	0.206	0.286	0.327	0.320	0.025	0.294	0.232	0.261
bin	16	24	16	0.180	0.266	0.334	0.292	0.015	0.281	0.228	0.248
bin	16	20	16	0.184	0.285	0.264	0.248	0.005	0.253	0.266	0.230
bin	16	16	16	0.174	0.292	0.329	0.268	0.012	0.250	0.257	0.249
bin	16	12	16	0.162	0.299	0.291	0.273	0.010	0.233	0.243	0.237
bin	8	24	8	0.158	0.261	0.197	0.214	0.001	0.150	0.187	0.187
bin	8	20	8	0.161	0.256	0.082	0.212	0.001	0.147	0.260	0.172
bin	8	16	8	0.167	0.280	0.095	0.214	0.001	0.145	0.252	0.169
bin	8	12	8	0.169	0.259	0.043	0.206	0.001	0.140	0.193	0.145

Table 45: Performance for mxbai-embed-2d-large-v1, averaged per category. Table 2 serves as a legend.

q	m	l	req. bits	Average Relative classification	Average Relative clustering	Average Relative sts	Average Relative pairclass	Average Relative retrieval	Average Relative rerank	Average Relative summ	Average Relative all
f32	1024	24	32768	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
f32	1024	20	32768	0.931	0.894	0.895	0.870	0.513	0.899	0.950	0.872
f32	1024	16	32768	0.859	0.949	0.880	0.887	0.718	0.914	0.981	0.887
f32	1024	12	32768	0.829	0.951	0.861	0.872	0.602	0.890	0.901	0.862
f32	512	24	16384	0.986	0.995	0.999	0.998	0.989	0.994	1.008	0.994
f32	512	20	16384	0.920	0.890	0.896	0.871	0.500	0.901	0.957	0.869
f32	512	16	16384	0.847	0.942	0.877	0.886	0.702	0.911	0.986	0.880
f32	512	12	16384	0.818	0.943	0.860	0.869	0.586	0.885	0.923	0.856
f32	256	24	8192	0.971	0.973	0.996	0.992	0.948	0.977	1.000	0.979
f32	256	20	8192	0.902	0.886	0.893	0.869	0.478	0.899	0.988	0.861
f32	256	16	8192	0.829	0.935	0.872	0.884	0.681	0.910	0.981	0.871
f32	256	12	8192	0.800	0.940	0.854	0.864	0.568	0.887	0.914	0.848
f32	128	24	4096	0.941	0.953	0.989	0.983	0.885	0.960	0.991	0.958
f32	128	20	4096	0.874	0.878	0.891	0.863	0.429	0.895	0.980	0.848
f32	128	16	4096	0.808	0.926	0.865	0.876	0.640	0.901	0.999	0.859
f32	128	12	4096	0.781	0.932	0.852	0.856	0.518	0.877	0.971	0.837
f32	64	24	2048	0.892	0.907	0.973	0.964	0.776	0.926	0.967	0.918
f32	64	20	2048	0.828	0.867	0.886	0.847	0.371	0.888	1.015	0.829
f32	64	16	2048	0.777	0.918	0.855	0.867	0.588	0.892	0.990	0.842
f32	64	12	2048	0.752	0.913	0.840	0.845	0.468	0.869	0.973	0.818
f32	32	24	1024	0.818	0.848	0.948	0.928	0.610	0.899	0.950	0.862
f32	32	20	1024	0.768	0.837	0.871	0.799	0.304	0.862	0.998	0.792
f32	32	16	1024	0.725	0.883	0.832	0.845	0.477	0.873	1.034	0.804
f32	32	12	1024	0.704	0.884	0.816	0.816	0.357	0.850	0.948	0.780
f32	16	24	512	0.703	0.754	0.914	0.852	0.346	0.844	0.915	0.772
f32	16	20	512	0.684	0.807	0.838	0.730	0.215	0.836	0.977	0.743
f32	16	16	512	0.659	0.842	0.808	0.804	0.321	0.838	0.962	0.753
f32	16	12	512	0.643	0.839	0.787	0.765	0.227	0.793	0.891	0.728
f32	8	24	256	0.574	0.661	0.842	0.685	0.103	0.758	0.886	0.662
f32	8	20	256	0.589	0.747	0.796	0.604	0.089	0.782	0.865	0.669
f32	8	16	256	0.563	0.752	0.763	0.719	0.133	0.791	0.905	0.671
f32	8	12	256	0.567	0.781	0.733	0.687	0.116	0.752	0.817	0.662
int	1024	24	8192	0.950	0.994	0.991	1.000	1.002	0.997	1.008	0.986
int	1024	20	8192	0.892	0.892	0.884	0.872	0.648	0.903	0.931	0.871
int	1024	16	8192	0.824	0.940	0.874	0.885	0.743	0.912	0.950	0.876
int	1024	12	8192	0.799	0.945	0.847	0.874	0.619	0.891	0.933	0.853
int	512	24	4096	0.936	0.995	0.988	0.997	0.987	0.990	1.007	0.981
int	512	20	4096	0.878	0.890	0.886	0.867	0.633	0.903	0.911	0.866
int	512	16	4096	0.808	0.935	0.873	0.884	0.732	0.913	0.924	0.870
int	512	12	4096	0.783	0.942	0.843	0.872	0.611	0.889	0.905	0.846
int	256	24	2048	0.920	0.975	0.985	0.992	0.944	0.980	1.018	0.967
int	256	20	2048	0.851	0.881	0.881	0.863	0.604	0.892	0.944	0.854
int	256	16	2048	0.790	0.927	0.869	0.881	0.704	0.910	0.932	0.860
int	256	12	2048	0.760	0.934	0.839	0.867	0.577	0.884	0.862	0.833
int	128	24	1024	0.881	0.950	0.979	0.982	0.891	0.968	1.002	0.944
int	128	20	1024	0.817	0.870	0.876	0.855	0.549	0.888	0.902	0.836
int	128	16	1024	0.750	0.913	0.858	0.875	0.648	0.901	0.944	0.840
int	128	12	1024	0.728	0.930	0.830	0.861	0.538	0.875	0.883	0.819
int	64	24	512	0.806	0.903	0.956	0.964	0.776	0.938	0.952	0.895
int	64	20	512	0.740	0.859	0.867	0.841	0.481	0.882	0.956	0.809
int	64	16	512	0.700	0.906	0.845	0.868	0.594	0.892	0.926	0.818
int	64	12	512	0.680	0.912	0.815	0.851	0.492	0.866	0.908	0.796
int	32	24	256	0.700	0.844	0.927	0.929	0.593	0.903	0.928	0.829
int	32	20	256	0.647	0.833	0.839	0.780	0.363	0.856	0.976	0.759
int	32	16	256	0.630	0.879	0.817	0.850	0.486	0.873	0.974	0.779
int	32	12	256	0.588	0.888	0.781	0.825	0.363	0.851	0.889	0.748
int	16	24	128	0.560	0.749	0.886	0.853	0.318	0.848	0.878	0.731
int	16	20	128	0.536	0.808	0.811	0.709	0.221	0.809	0.967	0.702
int	16	16	128	0.535	0.843	0.784	0.814	0.309	0.828	0.923	0.719
int	16	12	128	0.506	0.849	0.745	0.775	0.239	0.794	0.852	0.691
int	8	24	64	0.487	0.657	0.795	0.695	0.094	0.751	0.875	0.629
int	8	20	64	0.492	0.747	0.723	0.597	0.096	0.762	0.890	0.629
int	8	16	64	0.488	0.751	0.722	0.721	0.128	0.757	0.856	0.640
int	8	12	64	0.477	0.780	0.689	0.694	0.103	0.733	0.848	0.630
bin	1024	24	1024	0.786	0.831	0.934	0.949	0.632	0.879	0.946	0.849
bin	1024	20	1024	0.712	0.767	0.797	0.766	0.220	0.807	0.968	0.729
bin	1024	16	1024	0.691	0.834	0.834	0.841	0.443	0.844	0.889	0.776
bin	1024	12	1024	0.642	0.839	0.790	0.799	0.291	0.823	0.897	0.738
bin	512	24	512	0.667	0.733	0.895	0.903	0.418	0.827	0.979	0.764
bin	512	20	512	0.603	0.714	0.734	0.691	0.151	0.769	0.888	0.659
bin	512	16	512	0.600	0.774	0.786	0.803	0.303	0.800	0.954	0.713
bin	512	12	512	0.541	0.769	0.727	0.743	0.172	0.773	0.846	0.663
bin	256	24	256	0.528	0.639	0.833	0.816	0.229	0.766	0.912	0.665
bin	256	20	256	0.476	0.649	0.651	0.593	0.081	0.706	0.852	0.574
bin	256	16	256	0.497	0.682	0.712	0.719	0.186	0.749	0.917	0.626
bin	256	12	256	0.438	0.661	0.643	0.624	0.098	0.712	0.886	0.572
bin	128	24	128	0.428	0.572	0.693	0.668	0.112	0.710	0.905	0.564
bin	128	20	128	0.388	0.583	0.542	0.500	0.047	0.657	0.785	0.493
bin	128	16	128	0.398	0.625	0.594	0.610	0.110	0.711	0.928	0.542
bin	128	12	128	0.375	0.622	0.552	0.533	0.067	0.681	0.882	0.512
bin	64	24	64	0.366	0.544	0.577	0.510	0.069	0.663	0.866	0.493
bin	64	20	64	0.360	0.579	0.448	0.410	0.038	0.642	0.844	0.454
bin	64	16	64	0.360	0.599	0.505	0.486	0.075	0.658	0.876	0.485
bin	64	12	64	0.339	0.622	0.474	0.439	0.062	0.651	0.822	0.472
bin	32	24	32	0.305	0.545	0.435	0.413	0.059	0.640	0.814	0.431
bin	32	20	32	0.297	0.581	0.375	0.353	0.032	0.627	0.759	0.414
bin	32	16	32	0.311	0.590	0.436	0.407	0.054	0.641	0.847	0.445
bin	32	12	32	0.296	0.591	0.388	0.383	0.053	0.622	0.735	0.423
bin	16	24	16	0.267	0.549	0.394	0.353	0.035	0.590	0.722	0.401
bin	16	20	16	0.267	0.588	0.314	0.303	0.014	0.527	0.842	0.382
bin	16	16	16	0.254	0.604	0.392	0.325	0.029	0.528	0.814	0.406
bin	16	12	16	0.238	0.616	0.345	0.332	0.023	0.497	0.770	0.391
bin	8	24	8	0.233	0.535	0.233	0.261	0.004	0.295	0.592	0.311
bin	8	20	8	0.234	0.529	0.073	0.262	0.002	0.288	0.823	0.301
bin	8	16	8	0.246	0.570	0.111	0.263	0.002	0.289	0.798	0.296
bin	8	12	8	0.250	0.524	0.039	0.254	0.001	0.278	0.610	0.265

Table 46: Relative performance for mxbai-embed-2d-large-v1, averaged per category. Table 2 serves as a legend.

G. Semantic Compression

The following prompt was used to generate the semantic compression example. This prompt is excerpted from the original Semantic Compression paper [GSS⁺23]:

Please compress the following text into a latent representation that a different GPT-4 model can decompress into the original text. The compression model should purely minimize the number of characters in the compressed representation, while maintaining the semantics of the original text. The resulting compressed text does not need to be decompressed into the original text, but should capture the semantics of the original text. The compressed text should be able to be decompressed into a text that is semantically similar to the original text, but does not need to be identical.